

General notes on tool collection



- selection of tools useful for DH
- online/web & offline/desktop applications
- (mainly) open & proprietary software
- 'grown' and open collection (non-exhaustive and exemplary selection)
- different tools – different purposes

Tool collection

Code editors and IDEs

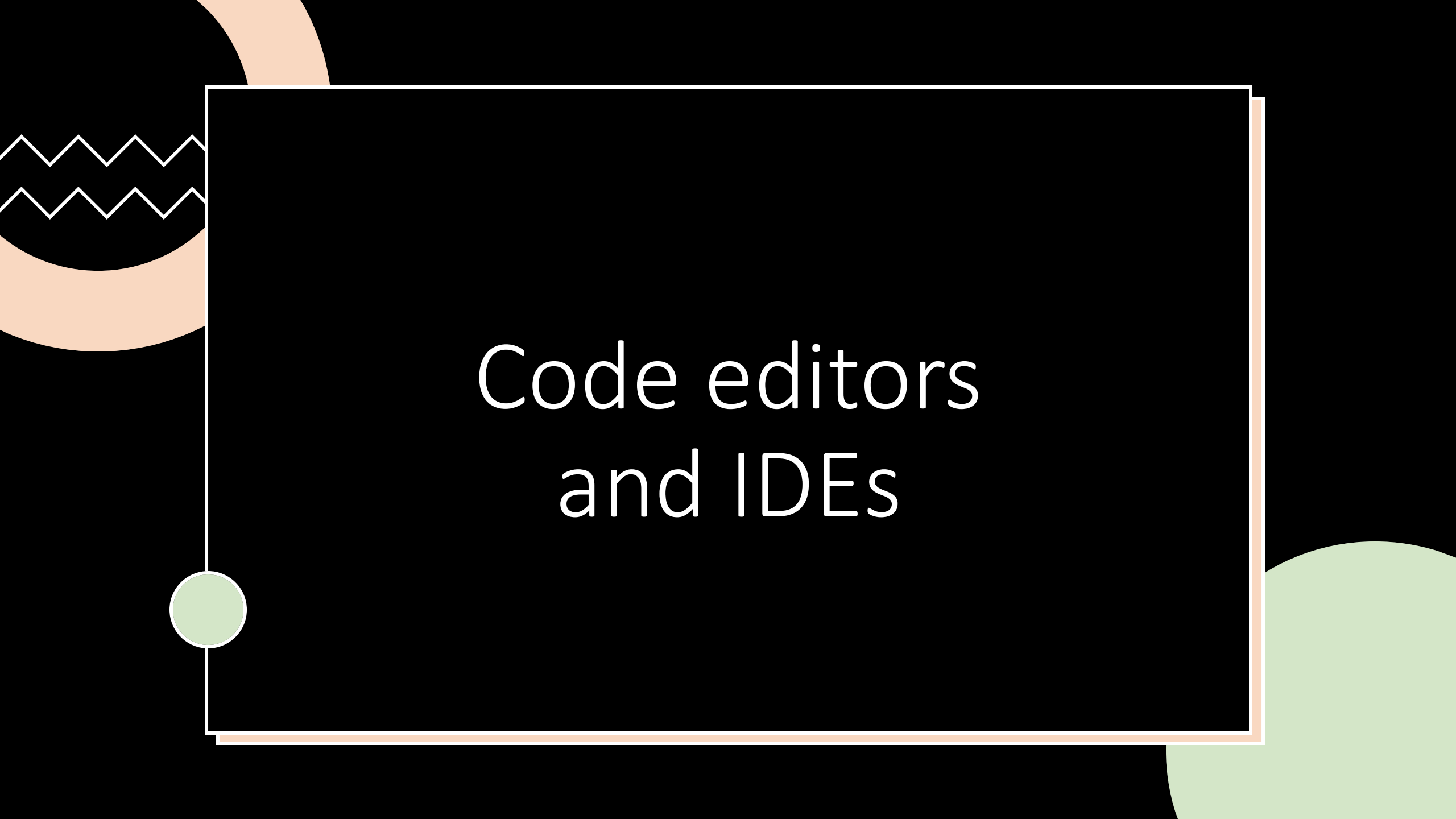
Automated Text Recognition

Annotation

Data Collection & Organisation

Data Analysis & Visualisation

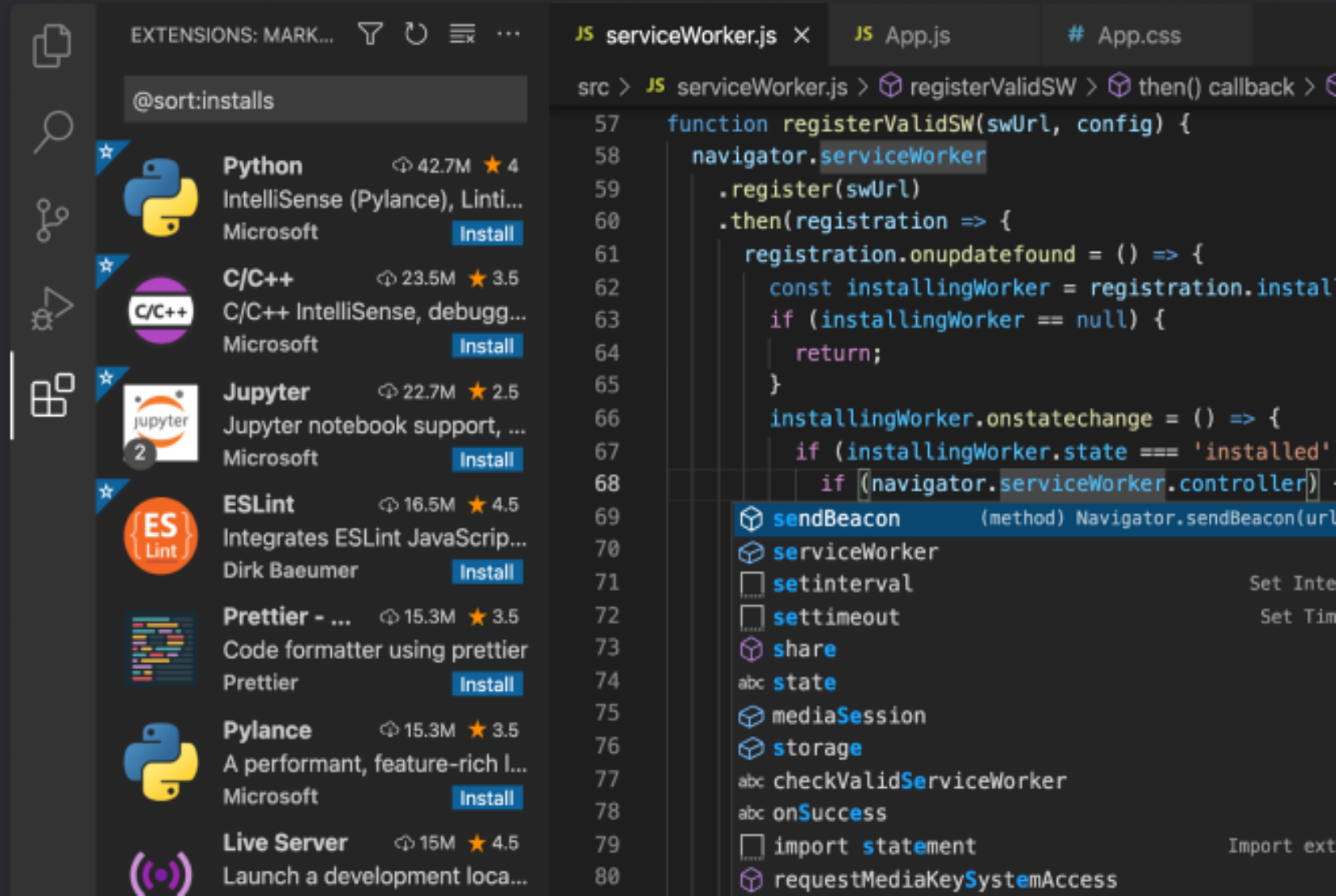
GIS, 3D Visualisation, and more...



Code editors and IDEs

Visual Studio Code <https://code.visualstudio.com/>

- Open Source Code Editor
- Extensions for various (programming) languages





<https://notepad-plus-plus.org>

- Free and open-source text/code editor
- "Notepad replacement", but can do much more!
- Only for Windows

```
C:\Users\mcarlon\GitHub\Teaching_CBS4DH\exercise\git\hands-on_1.md - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
hands-on_1.md
1  # Hands-on instructions
2
3  ## 1. Configuring git and creating a new repository from the command line
4
5  + Open a terminal (UNIX shell / bash is assumed).
6
7  + Let's make sure that the 'git' command is available: '$ git --version'
8
9  + One-time Git setup (set author info, text editor, line endings):
10 + '$ git config --global user.name "<your name>"'
11 + '$ git config --global user.email "<username>@users.noreply.github.com"'
12 + '$ git config --global core.editor "nano -w"'
13 + '$ git config --global core.autocrlf input' for Mac/Linux, 'true' for Windows
14 + Check settings: '$ git config --global --list' (or '$ cat ~/.gitconfig')
15
16 + _Notes_:
17 + The email you set here will potentially be visible to everyone on the internet, which is why you might want to use the scheme above.
18 + By default, the text editor is used for writing commit messages, but you can also use 'git commit -m '<message>'. Set your preferred editor.
19 + These (and all other config settings) can also be set on a per-repository level by omitting the '--global' flag. Per-repository settings are stored in a file named '.gitconfig' in the repository directory.
20 + For more information on line endings (and Git config in general), see <https://www.git-scm.com/book/en/v2/Customizing-Git-Git-Configuration>
21
22 + Let's create a brand-new repository from scratch:
23 + Using the 'cd' command, navigate to the parent directory of where you want to create your new repository (e.g. '~/Documents' if you want to create it in your home directory).
24 + Create a new folder there and 'cd' into it: '$ mkdir my_first_repo', '$ cd my_first_repo'
25 + Initialize a new repository: '$ git init'
26
27 + Take a quick look at the newly created '.git' directory:
28 + List contents of current directory ('-a' option shows hidden items): '$ ls -a'
29 + List contents of '.git': '$ ls .git'
30 + View '.git/config' file: '$ cat .git/config'
31
32
33 ## 2. Working with files and commits from the command line
34
35 + We continue with the fresh repository created in the first hands-on session.
36
37 + Create and save a new text file with some text (traditionally "hello world") in it: '$ echo 'hello world' > hello.txt' (or use your favorite text editor)
38
39 + '$ git status' should list the newly created file under "Untracked files".
40
```

Code editors or IDEs?

- IDE = Integrated Development Environment
- More tools for debugging and compiling
- Usually for a specific language

More info: <https://blog.jetbrains.com/webstorm/2024/03/ides-vs-code-editors>



Several IDEs
([IntelliJ](#), [PyCharm](#) etc.)



[Visual Studio](#)
(Win, Mac, Linux)



[Xcode](#)
(Mac only)



[Eclipse IDE](#) for Java
(Win, Mac, Linux)

GitHub Student Developer Pack

Learn to ship software like a pro. There's no substitute for hands-on experience. But for most students, real world tools can be cost-prohibitive. That's why we created the GitHub Student Developer Pack with some of our partners and friends.

Sign up for Student Developer Pack

Love the pack? Spread the word

<https://education.github.com/pack>

Experiences

Discover the best ways to use pack offers with Experiences. Experiences are curated bundles of pack partner products, GitHub tools, and other resources that are designed for you learn new skills and make the most out of the Student Developer Pack and your journey in Global Campus.



Tools

AI Prompting & Technical Writing

This Experience offers the tools, top tips and resources you'll need to get started becoming a technical wizard with words.

Offers in this bundle



[Learn more about AI Prompting & Technical Writing >](#)



Tools

Security and Monitoring

Explore industry standard tools that keep your code safe, your permissions secure and your infrastructure monitored so that you can confidently scale your brilliant ideas.

Offers in this bundle



[Learn more about Security and Monitoring >](#)



Primer

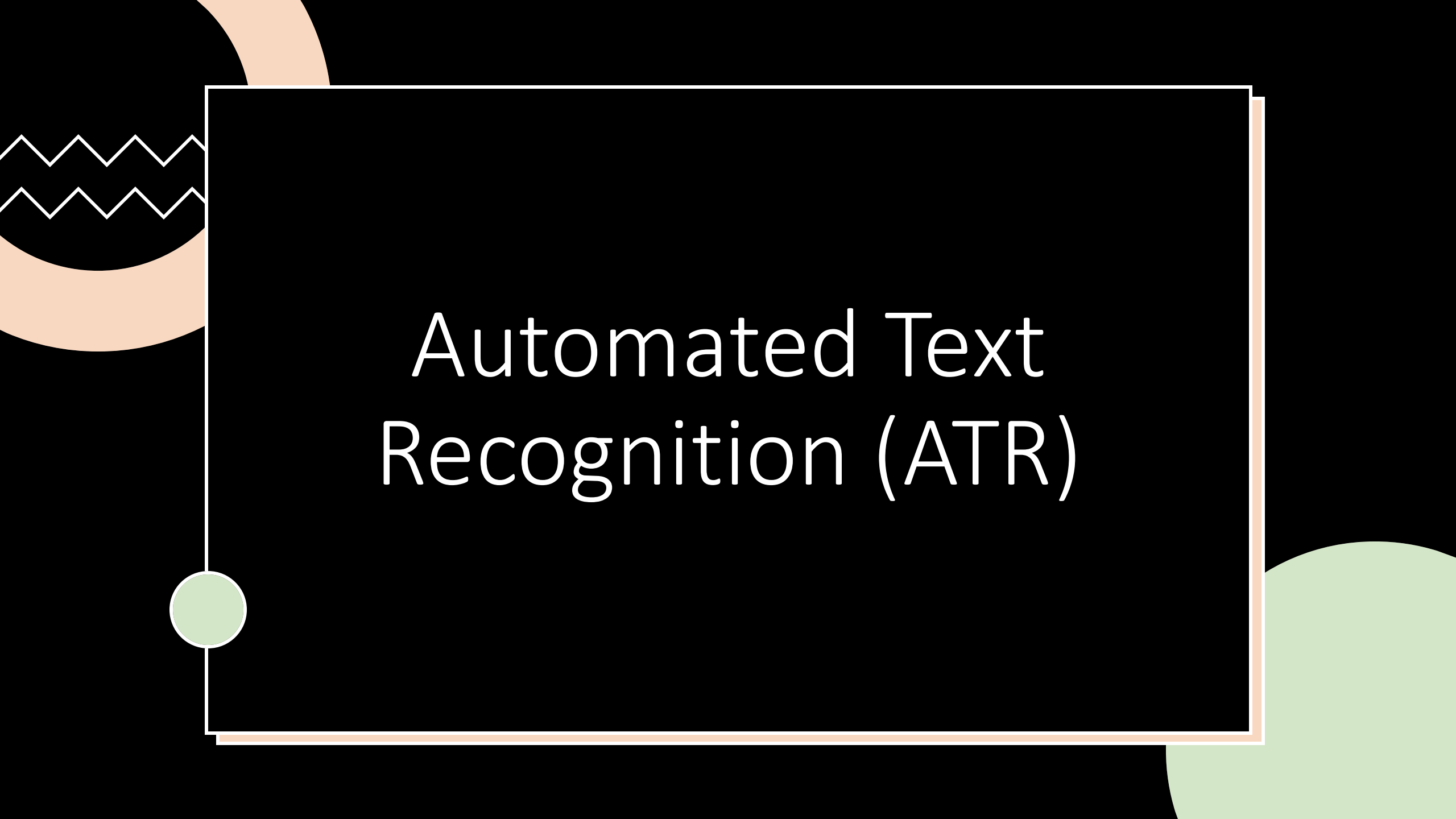
Primer: Copilot

What is Copilot and how do you get started using it? Use this Experience as a Primer on Copilot. We share tutorials, guides and templates to learn how to use AI in your development process.

Offers in this bundle



[Learn more about Primer: Copilot >](#)

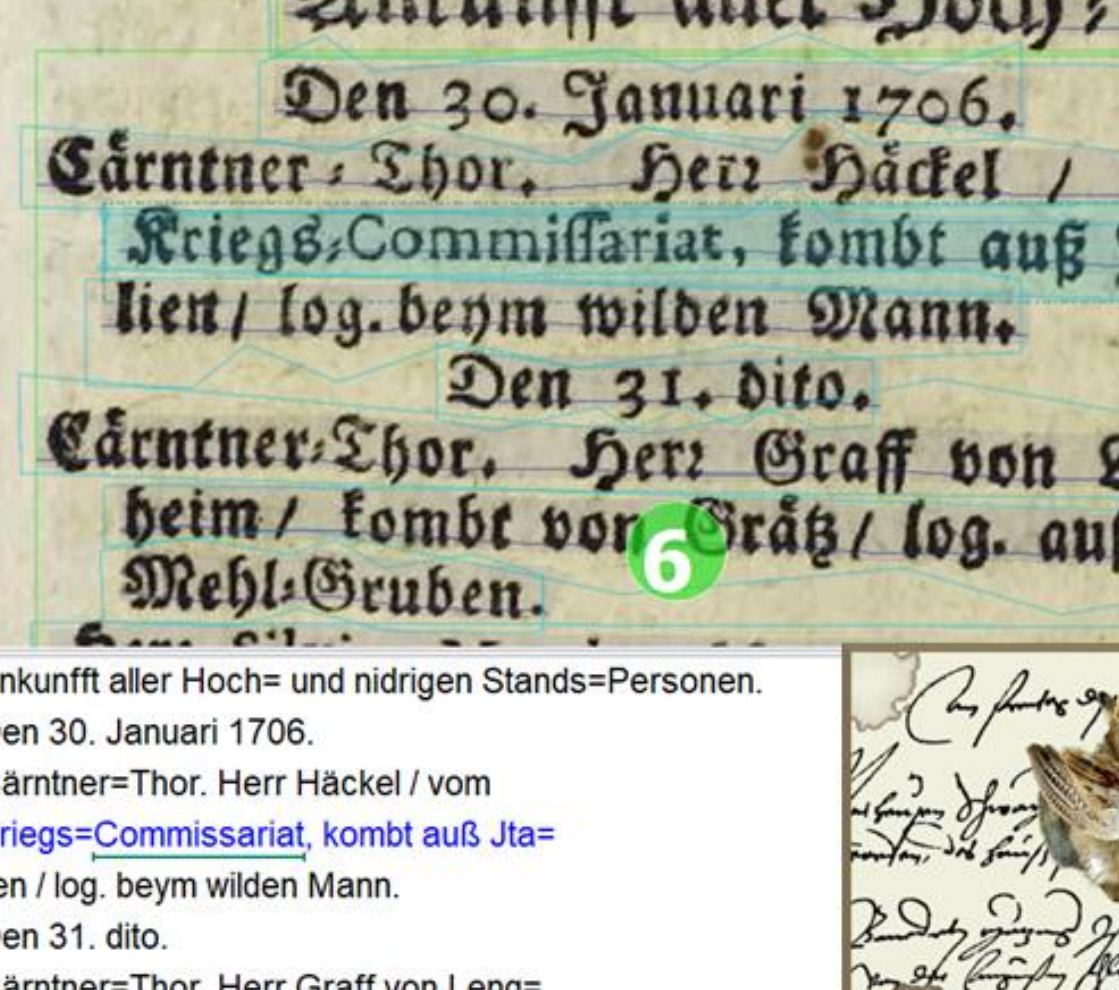


Automated Text Recognition (ATR)



<https://readcoop.eu/de/transkribus/>

- Layout and text recognition
- HTR = Handwritten Text Recognition, can be used for handwritten and printed text
- Online and local version available (Expert Client recommended)

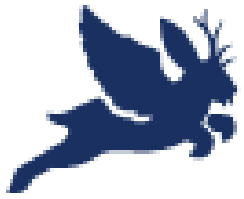


TR
L
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6

Ankunfft aller Hoch- und
Den 30. Januari 1706.
Cärntner-Thor. Herr Häckel / vom
Kriegs-Commissariat, kombt auß Ita-
lien / log. beym wilden Mann.
Den 31. dito.
Cärntner-Thor. Herr Graff von Leng-
heim / kombt von Grätz / log. auf der
Mehl-Gruben.

5-1 Ankunfft aller Hoch= und nidrigen Stands=Personen.
6-1 Den 30. Januari 1706.
6-2 Cärntner=Thor. Herr Häckel / vom
6-3 Kriegs=Commissariat, kombt auß Jta=
6-4 lien / log. beym wilden Mann.
6-5 Den 31. dito.
6-6 Cärntner=Thor. Herr Graff von Leng=
6-7 heim / kombt von Grätz / log. auf der
6-8 Mehl=Gruben.

Transkribus



Transkribus[®]

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Sprache:

Deutsch



Handschrift

Druckschrift

KI-Modell: The German Giant I ⓘ

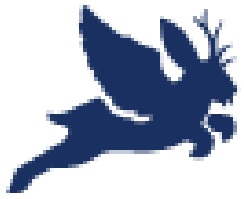
Ziehen Sie hier ein Bild rein
oder

Datei auswählen...

PNG oder JPG bis zu 10 Mb

Mit dem Hochladen eines Bildes akzeptieren Sie unsere [Allgemeinen Geschäftsbedingungen](#) und unsere [Datenschutzrichtlinie](#).





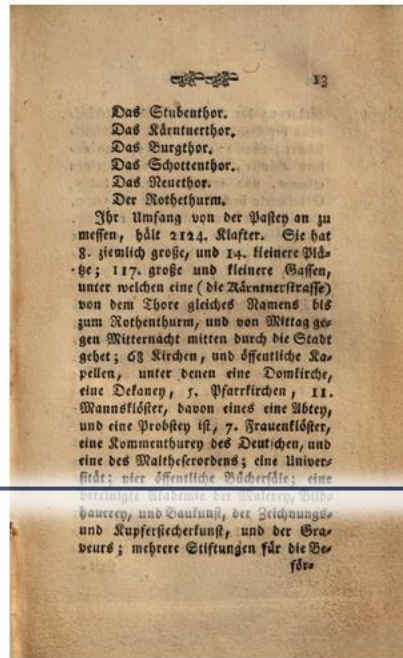
Transkribus®

[Transkribus](#)[ScanTent](#)[read&search](#)[Über](#)[Ressourcen](#)[Pläne & Preise](#)[Anmelden](#)[App](#)

Sprache: Deutsch

Handschrift Druckschrift

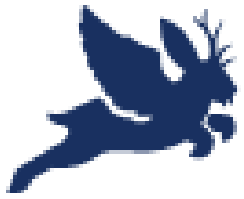
KI-Modell: The German Giant I ⓘ



Transkribieren..

Mit dem Hochladen eines Bildes akzeptieren Sie unsere Allgemeinen Geschäftsbedingungen und unsere Datenschutzrichtlinie.



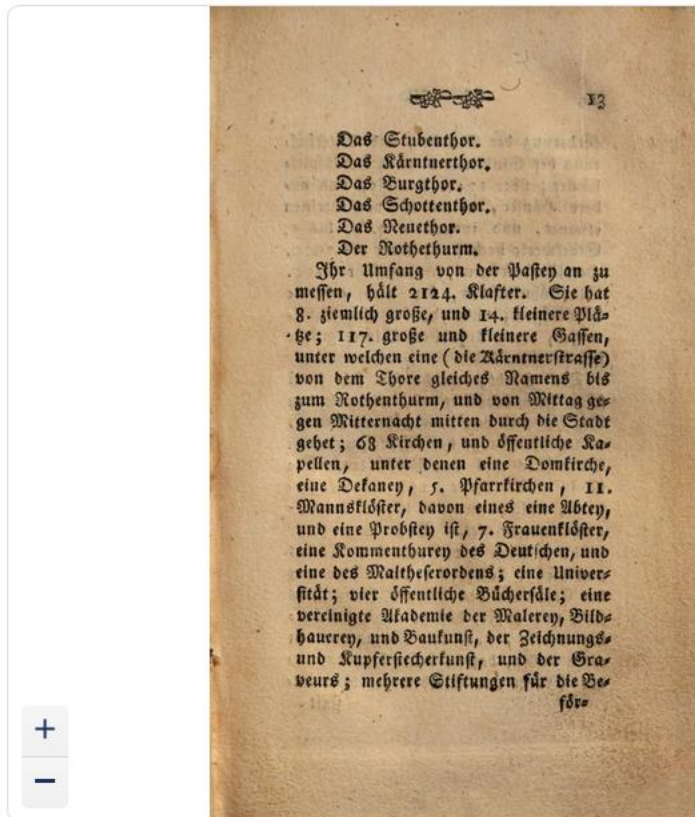


Sprache: Deutsch

Handschrift

Druckschrift

KI-Modell: The German Giant I ⓘ



73

Das Stubenthor.

Das Kärntnerthor.

Das Burgthor.

Das Schottenthor.

Das Neuethor.

Der Rothethurm.

Ihr, Umfang von der Pastey an zu messen, hält 2124. Klafter. Sie hat 8. ziemlich große, und 14. kleinere Plätze; 117. große und kleinere Gassen, unter welchen eine (die Kärntnerstrasse) von dem Thore gleiches Namens bis

Anderes Bild hochladen

PDF

Doc

In die Zwischenablage kopieren





Transkribus®

Wiener
Zeitung

paragraph

list

mit 200., 3. andere mit 300., und 1. desto mit 500. Ducaten abgeführt worden.

Und zumalen die dormalige Pachtung des sogenannten Lotto di Genova in andere Hände gekommen, so wird hiemit zu wissen gemacht, daß alldiejenige, welche etwann noch ein-oder andere Gewinne an sich zu bringen haben, sich in Zeit von 6. Monaten gegen Probenung ihrer Originalloose gehörig melden sollen, wo im widrigen nach Verfließung dieser Zeit, nach Vorschrift der Allerhöchsten Verordnung derselben Lose für null und nichtig angesehen, und ohne einigen Ausnahm oder Vorbehalt für ungültig und kraftlos erklärt werden sollen.

Liste deren Verstorbenen zu Wien im und vor der Stadt

Den 11. Mart. In der Stadt.
Der Woldebeck. Hr. Joh. Bapt. Rejon, Churpfälz. Hofrath, und R. R. Hofkriegs- rathsf. agent, im Rezenfurgerhof, alt 69. J.
Dem Hrn. Maximil. Panholzer, R. R. Kammerthürh. f. R. Elisabeth, im K. Burgl. H. in der Naglergaf., alt 6. J.
Fr. Maria Eva Kaufsch. Wittwe, im Seelauff. H. im tief. Grab., alt 72. J.
Jacob Kiezer, Herrschaftl. Portier, im Volkert. H. am Kohlmarkt, alt 47. J.
Theres. Wanklin, led., im Kupferschmiedl. H. im tief. Grab., alt 38. J.
Cathar. Haberlerin, ledig, im Baberl. H. in der Wolfseil, alt 80. J.

Vor der Stadt.
Dem Joh. Mich. Schuf. Stadt- musico, f. R. Theres., beim geld. 1. am Epistalb., 5. v. J.

Den 12. Mart.
In der Stadt.
Dem Hrn. Domin. Ant. Pegari, Bürgerl. Handelsmann, f. R. Anna B. im Waghingerl. H. am alt. Baurenmarkt, alt 27. J.

Vor der Stadt.
Anna Waprin, ledig, beim weif. Stern in der Josephst., alt 61. J.
Dem Adam Zeit, arm. im Grundspit., außer Maria- hülfe, alt 80. J.
Dem Paul Kober, Led., f. R. Ward., im Pinapfl. H. auf der Landstr., alt 12. J.
Franz Wagner, Schmaltztrag., im Wagensenters. H. in der Leopoldst., alt 68. J.
Summa 5. Person. darunter 1. Kind.

Den 13. Mart. In der Stadt.
Dem Hochedelgeb. Hrn. Math. von Weittinger, Ihrer R. R. Majest. Hofrath, und Referendar. bey dem Hofkriegs- rath, f. R. Carolina, geb. v. Wundershofen, welche sich erfallen, ist bey dem gr. Nöf. in der Schulerstr. alldayom R. R. Stadt- und Landgerichte beschaet worden, alt 11. J.
Der Woldebeck. Hr. August. Nuregio, weltl. Prief. im Albrechtsburgl. H. im Berg- hof, 76. J.
Dem For. Reißer, Bürgerl. Schuhmach., f. W. Eva, im Schellerl. H. im Hagenst., alt 66. J.
Dem Joh. Christian B. in, Bürgerl. Hand- handl., f. R. Joh. Christian, im Kernl. H. am Salzgr., alt 1. J.
Dem Math. Trombeter, gen. R. R. Trab., f. W. Eva, beim gold. A. B. C. in der Himmelst. gaf., alt 69. J.
Dem Joh. Mich. Frank, Herrschaftl. Hefebüch., f. W. Magdal., im Caffee- fiederl. H. auf der Schottenbalken, alt 26. J.
Ward. Kreglin, led. W., im Lill. H. in der Eisen- gerstr., alt 29. J.

Vor der Stadt.
Der Woldebeck. Hr. Joh. v. Pirelli, gen. R. R. Obrist- wacht- weif., f. R. Kallingerl. H. in der Josephst., alt 50. J.

heading

catch-word

Field Model

ID: 56638



Private Model

German Newspapers 18th Century - WrDiarium M3

by NinaClaudia.Rastinger@oeaw.ac.at

Nov 16, 2023

Trained

Tags

paragraph, heading, header, list, main-title, imprint, count-date, catch-word, figure, seperator-double, seperator-single, signature

% Accuracy

81.81%

Trained on

handwritten

Edit

Show Description

Transkribus Scholarship Programme

The Transkribus Scholarship Programme is designed to encourage and support future scholars as this is one of our core values. With this programme we want to facilitate the research in historical documents.

The programme is open to students and teachers from any programme or discipline. Through this global programme you can get free credits to use for the Handwritten Text Recognition technology in Transkribus. Transkribus Scholarships are granted based on projects, this means that you need to apply with a specific project that you plan to spend the credits of your Scholarship on.



<https://readcoop.eu/de/transkribus/scholarship/>



eScriptorium

eScriptorium: A Digital Text Production Pipeline for Print and Handwritten Texts using machine learning techniques.



Automatic Transcription

Apply OCR/HTR to images of printed and handwritten documents using shared open models.



Manual transcription

Make use of an ergonomic user interface leveraging modern browser technology to edit segmentations and transcriptions.



Train Models

Create new models or finetune existing ones to improve automatic recognition.



Import/Export

Import and export models and texts transcriptions in a variety of formats. Access data through a full REST API.

<https://escriptorium.readthedocs.io/en/latest/>

OCR4all

Optical Character Recognition (and more) for everyone

Setup guide, user guide, developer documentation and more.



<https://www.ocr4all.org/>

Get Started

Fully free and open-source

OCR4all is and will stay completely free and open-source. No subscriptions, paywalled features or private code.

Flexible applicable

From the high-quality processing of challenging manuscripts to the mass full-text recognition of printings

Powerful layout and text annotation included

Manually annotate, correct or compare layout and text elements using the powerful LAREX editor.

OCR-D compatible

All future versions of OCR4all are built to be fully compatible with the OCR-D ecosystem

Designed with usability in mind

Create complex OCR workflows through the UI without the need of interacting with code or command line interfaces.

Easy cross-platform deployment

Docker and a single command are all it takes to get OCR4all up and running, regardless of your operating system.

Tutorials

- Transkribus Tutorial Playlist on Youtube: <https://www.youtube.com/playlist?list=PL7UbQtd4qlhIMP1KfdjGW3C-KXTxw4KYb>
- Transkribus Helpcenter: <https://help.transkribus.org/getting-started>
- ForTEXT Tutorial for OCR4all: <https://fortext.net/tools/tools/ocr4all>



Annotation



Text Encoding Initiative

W.H.Auden

77 St Mark's Place
New York City 3
New York I0003
U.S.A.

Air-Mail

Die Baronin
Stella Musulin

Fridau
3200 Obergrafendorf
N.Ö.
Austria



amp-transcript__0004.xml X

TEI

```
157      </surface>
158      </facsimile>
159      <text type="letter" hand="#typed">
160        <body>
161          <div type="transcription" xml:id="transcription_amp_0004" xml:lang="en">
162            <div type="envelope">
163              <pb facs="https://iiif.acdh.oeaw.ac.at/amp/amp_0019/" type="envelope">
164                <ab>
                  <rs type="person" ref="#amp_person_1">W.H.Auden</rs><lb/>
                  <address>
                    <rs type="place" ref="#amp_place_id_29">77 St Mark's Place</rs>
                    New York City 3<lb/>
                    New York I0003<lb/>
                    U.S.A.</rs>
                  </address>
                </ab>
                <ab>
                  <hi rend="underline">Air-Mail</hi>
                </ab>
                <ab xml:lang="de">
                  <rs type="person" ref="#amp_person_2">Die Baronin</rs>
                  <space quantity="3" unit="char"/>Stella Musulin</rs><lb/>
                  <address>
                    <space quantity="5" unit="char"/><rs type="place" ref="#amp_place_id_30">Fridau</rs>
                    <space quantity="6" unit="char"/>3200 Obergrafendorf</rs>
                    <space quantity="8" unit="char"/>N.Ö.</rs>
                    <space quantity="9" unit="char"/><foreign xml:lang="en">Austria</foreign>
                  </address>
                </ab>
```

<https://amp.acdh.oeaw.ac.at/>

Semantic Annotation without the pointy brackets

Work on texts and images. Identify and mark named entities. Use your data in other tools or connect to other data on the Web. Without the need to learn code.



Shared Collections

Upload texts and images. Build your own collections. Invite others to collaborate.



Collaborative Annotation

Create notes. Tag. Discuss. Recogito tracks provenance and maintains version history.



Semantics Included

Tag places, people and events and reveal the relations between them.



Open Data

Export your data into other tools. Make the results of your work more open, visible and re-usable.



<https://catma.de/>

= Computer Assisted Text Markup and Analysis

CATMA 6.0.0 Edgar Allan Poe

THE TELL-TALE HEART-2 x

Edgar Allan Poe

TELL-TALE HEART

TRUE! --nervous --very, very dreadfully nervous I had been and am; but why will you say that I am mad? The disease had sharpened my senses --not destroyed --not dulled them. Above all was the sense of hearing acute. I heard all things in the heaven and in the earth. I heard many things in hell. How, then, am I mad? Hearken! and observe how healthily --how calmly I can tell you the whole story.

It is impossible to say how first the idea entered my brain; but once conceived, it haunted me day and night. Object there was none. Passion there was none. I loved the old man. He had never wronged me. He had never given me insult. For his gold I had no desire. I think it was his eye! yes, it was this! He had the eye of a vulture --a pale blue eye, with a film over it. Whenever it fell upon me, my blood ran cold; and so by degrees --very gradually --I made up my mind to take the life of the old man, and thus rid myself of the eye forever.

Collection currently being edited
Tell-Tale Heart Jan

Tagsets

Tagsets	Tags
character	opponent
	hero(ine)
	hero
narrator	attitude
	questionable morals
	questionable factual claims
	style
	address to the reader
	repetition
	exclamation
	parallelism

Project
Tags
Annotate
Analyze

21 % page size



+ GitMA

(<https://catma.de/documentation/access-your-project-data/git-access/gitma/>)

= Python library to process, analyse and manipulate CATMA annotations



Data Collection and Organisation

BeautifulSoup

<https://beautiful-soup-4.readthedocs.io/en/latest/>

– “Python library for pulling data out of HTML and XML files”

```
from bs4 import BeautifulSoup
soup = BeautifulSoup(html_doc, 'html.parser')

print(soup.prettify())
# <html>
# <head>
# <title>
#   The Dormouse's story
# </title>
# </head>
# <body>
# <p class="title">
# <b>
#   The Dormouse's story
# </b>
# </p>
# <p class="story">
#   Once upon a time there were three little sisters; and their names were
#   <a class="sister" href="http://example.com/elsie" id="Link1">
```



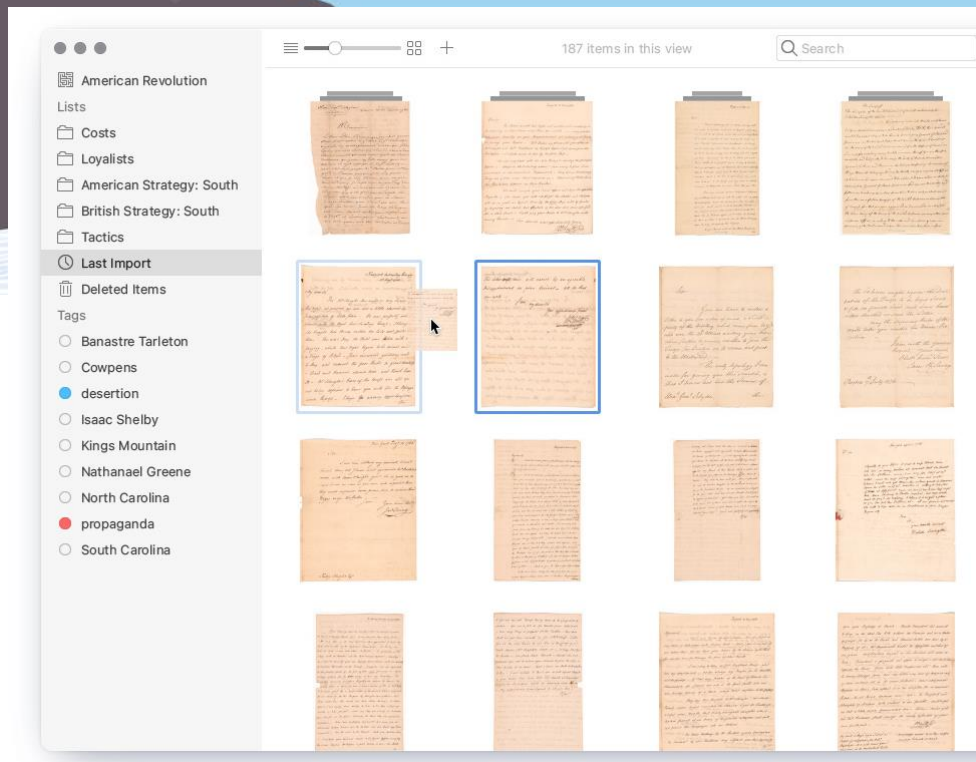


Explore your research photos

Download Tropy for **Windows (x64)**



What's new in version 1.16.2



Organisation, metadata enrichment
and annotation of images

<https://tropy.org/>



OpenRefine

OpenRefine is a powerful free, open source tool for working with messy data: cleaning it; transforming it from one format into another; and extending it with web services and external data.

[Download](#)<https://openrefine.org/>

Main features



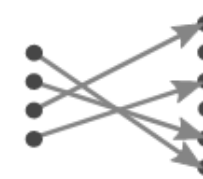
Faceting

Drill through large datasets using facets and apply operations on filtered views of your dataset.



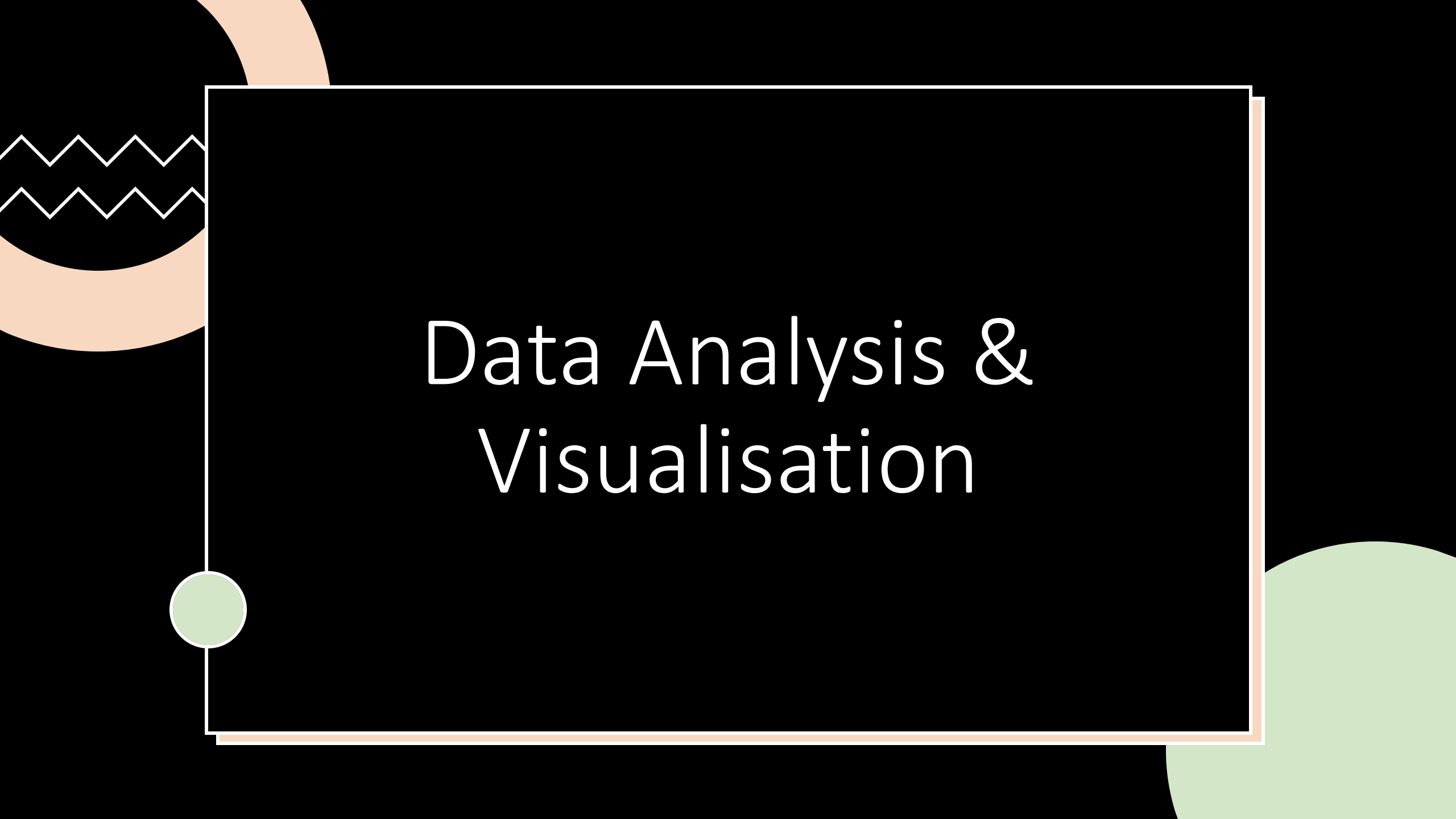
Clustering

Fix inconsistencies by merging similar values thanks to powerful heuristics.



Reconciliation

Match your dataset to external databases via reconciliation services.



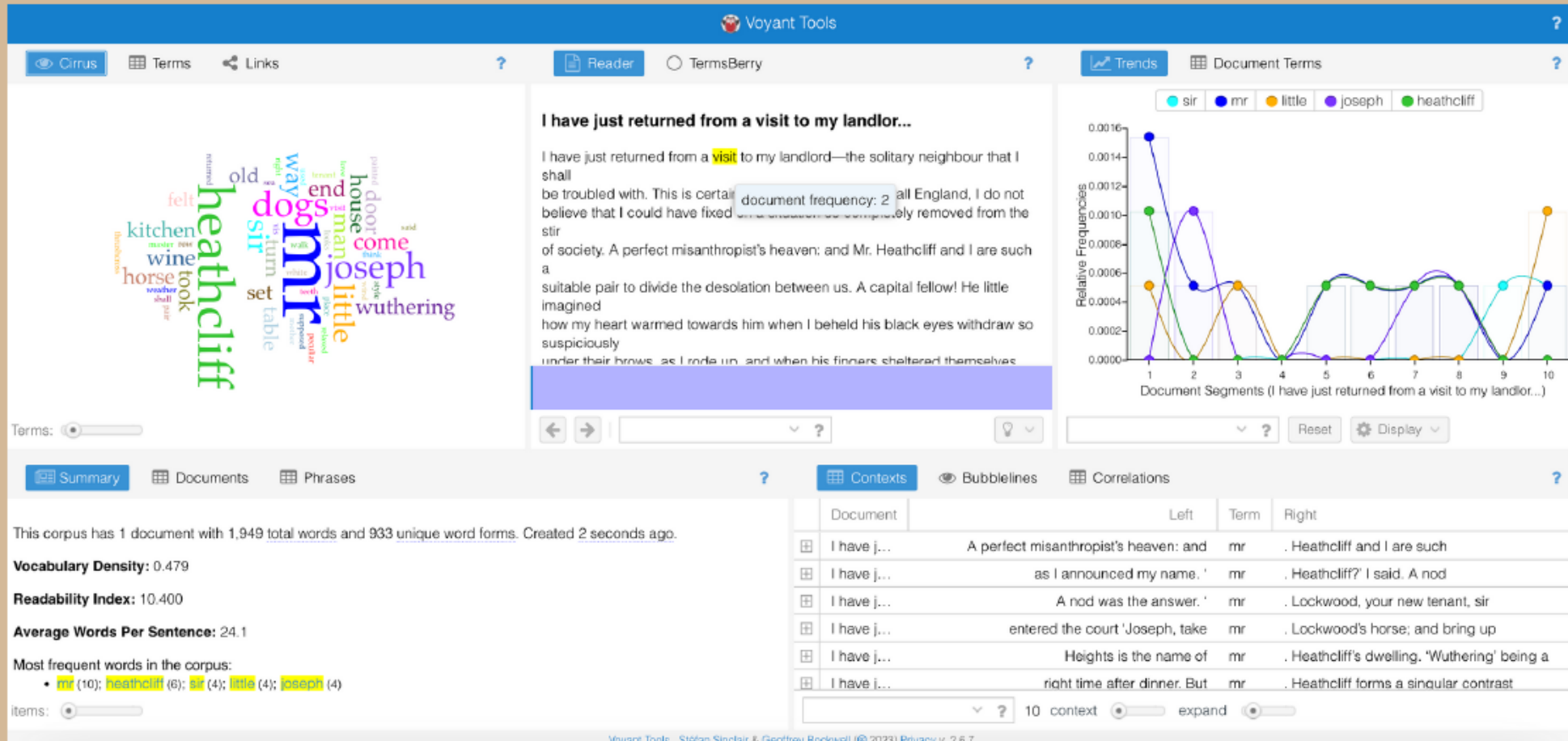
Data Analysis & Visualisation

VOYANT TOOLS

<https://voyant-tools.org/>



text
analysis
tool





Text hinzufügen



Geben Sie eine oder mehrere URLs in separate Zeilen ein oder fügen Sie einen Volltext ein.

 Öffnen

Hochladen

 Anzeigen

*Voyant Tools ist eine webbasierte Lese- und Analyseumgebung für digitale Texte..
Deutsche Übersetzung von Jan Theurl*

VOYANT

see through your text

Öffnen

Ein Korpus wählen:

Shakespeare's Plays
Austen's Novels
Mary Shelley's Frankenstein

Text hinzufügen

Geben Sie eine oder mehrere URLs

Öffnen

Hochladen

Anzeigen

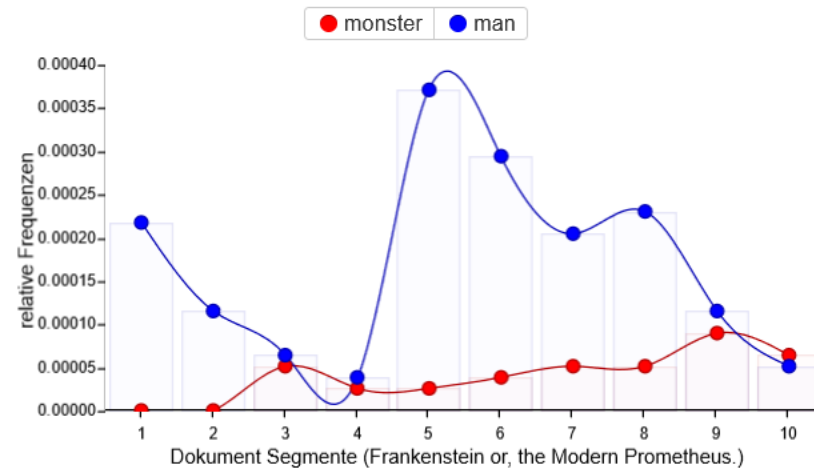
*Voyant Tools ist eine webbasierte Lese- und Analyseumgebung für digitale Texte..
Deutsche Übersetzung von Jan Theurl*



Introduction

The Publishers of the Standard Novels, in selecting Frankenstein for one of their series, expressed a wish that I should furnish them with some account of the origin of the story. I am the more willing to comply, because I shall thus give a general answer to the question, so very frequently asked me — How I, then a young girl, came to think of, and to dilate upon, so very hideous an idea? It is true that I am very averse to bringing myself forward in print; but as my account will only appear as an appendage to a former production, and as it will be confined to such topics as have connection with my authorship alone, I can scarcely accuse myself of a personal intrusion.

It is not singular that, as the daughter of two persons of distinguished literary celebritv. I should verv early in life have thought of writing. As a child I scribbled: and



monster x man x Zurücksetzen Darstellung

Dieses Korpus enthält 1 Dokument mit insgesamt 78,030 verschiedenen/ einzigartigen Wortformen. Erstellt vor ca. 2 Jahren.

Wortschatzdichte: 0.094

Readability Index: 11.653

Durchschnittliche Wortanzahl pro Satz: 22.9

Die häufigsten Begriffe im Korpus

- man (133); life (124); father (113); shall (109); eyes (107)

Elemente

Dokument	links	Begriff	rechts
Franken...	think that I was the	monster	that he said I was
Franken...	in daily fear lest the	monster	whom I had created should
Franken...	of your remaining friends. Abhorred	monster	! Fiend that thou art! The
Franken...	I was in reality the	monster	that I am, I was
Franken...	me. Was I, then, a	monster	, a blot upon the earth

monster x 31 Kontext erweitern



AntConc

<https://www.laurenceanthony.net/software/antconc/>

corpus
analysis
tool

File Edit Settings Help

Target Corpus KWIC Plot File View Cluster N-Gram Collocate Word Keyword Wordcloud

Name: AmE06_Learned **Total Hits: 87** Page Size 100 hits 1 to 87 of 87 hits

Files: 80 Tokens: 161469

File	Left Context	Hit	Right Context
1 AmE...	It is, however, prompted by the need to place the	process	of taking moments in context. Moments of the Distribution
2 AmE...	f online distance education. Successful online teaching is a	process	of taking our very best practices in the classroom
3 AmE...	in their parents' homes. The findings demonstrate that the	process	of assimilation was not uniform for all groups. Some
4 AmE...	with the Communist Party of Indonesia, which was in the	process	of being eliminated by Soeharto's New Order government.
5 AmE...	acred texts," the canon of modern children's literature. The	process	of creating or augmenting professional identity relied partia
6 AmE...	eyes, you lack that protein. Now scientists are in the	process	of figuring out which proteins are coded for by
7 AmE...	(2004), Donlan and Martin (2004), and Pysek et al. (2004).	Process	of invasion At one level, the issue of invasive
8 AmE...	or and the other participants is formed, through which the	process	of knowledge acquisition is collaboratively created. (See Ch
9 AmE...	express an inference. An inference , in turn, is a mental	process	of linking propositions by offering support to one propositi
10 AmE...	een theoretically appropriate for explaining the adaptation	process	of newcomers who arrived in America in the early 20
11 AmE...	motoric instructions, either unmodified or modified by the	process	of overlap. We postulate a parallel language-specific proces
12 AmE...	asured confounding is accounting for the findings, as the	process	of randomization makes the mathematical probability of su
13 AmE...	nt residue of mantle differentiation including the on-going	process	of seafloor spreading and building of island arcs. It
14 AmE...	ikszentmihalyi's (1990) concept of "flow" is a more general	process	of self-actualization. In flow, a person's tasks
15 AmE...	a voice of "several strengths." Her voice thereby enacts a	process	of the black community speaking to itself and explores
16 AmF...	is not. it is that the critical thinker takes the	process	of thinking seriously, consciously attends to that process an

Search Query Words Case Regexp Results Set All hits Context Size 10 token(s)

process Start Adv Search

Sort Options Sort to right Sort 1 1R Sort 2 2R Sort 3 3R Order by freq

Progress 100%

Time taken (creating kwic results): 0.3314 sec



<https://www.sketchengine.eu/nosketch-engine/>

NoSketch Engine



DASHBOARD

Susanne



SUSANNE

CORPUS INFO



Concordance

Examples of use in context



Parallel Concordance

Translation search



Wordlist

Frequency list



Keywords

Terminology extraction



Trends

Diachronic analysis, neologisms



Text type analysis

Statistics of the whole corpus



You are using NoSketch Engine. These tools are only available in Sketch Engine

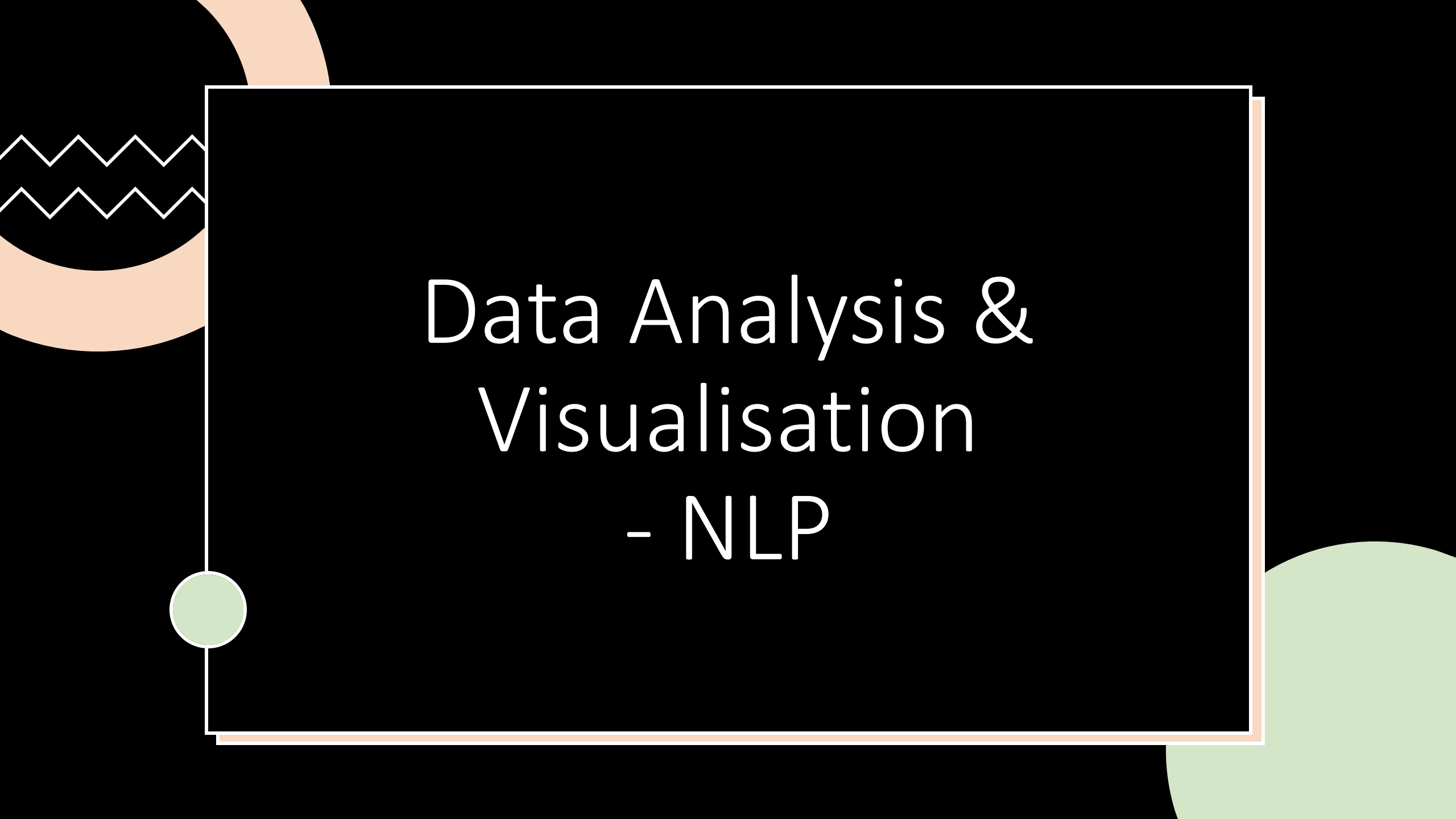
FREE CORPORA

English Susanne 128,998 words

OPEN

Tutorials

- Corpus Analysis with AntConc: <https://programminghistorian.org/en/lessons/corpus-analysis-with-antconc>
- Getting started with Voyant Tools: <https://voyant-tools.org/docs/#!/guide/start>



Data Analysis & Visualisation - NLP

NLP = Natural Language Processing

Exemplary NLP areas and tasks:

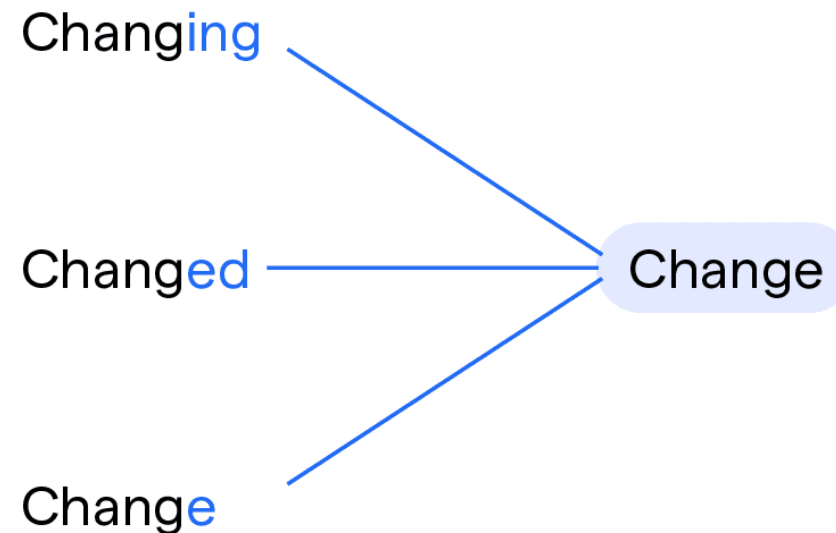
- Tokenization



NLP = Natural Language Processing

Exemplary NLP areas and tasks:

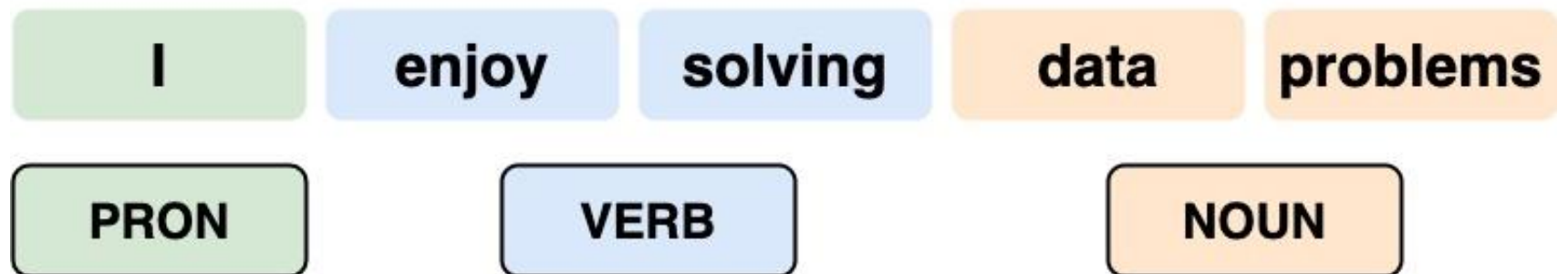
- Tokenization
- Lemmatization



NLP = Natural Language Processing

Exemplary NLP areas and tasks:

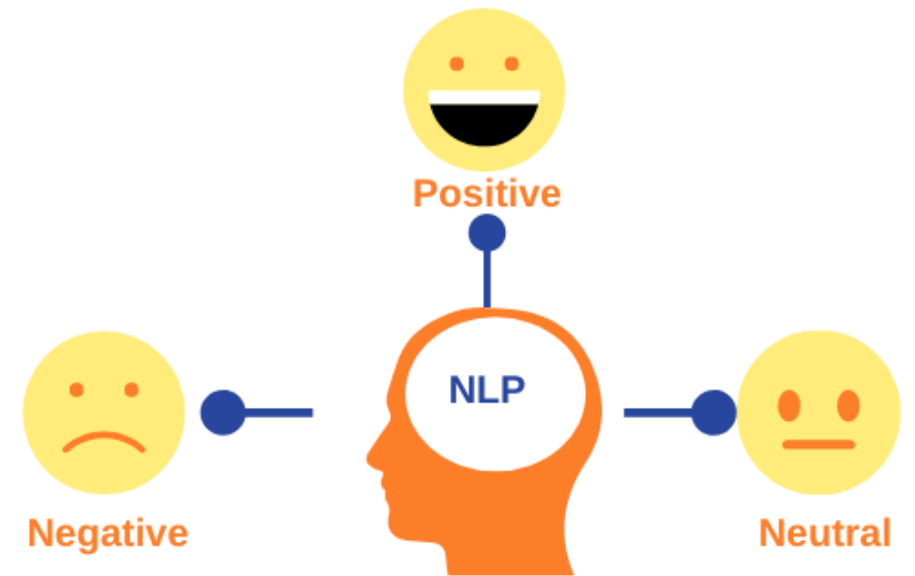
- Tokenization
- Lemmatization
- Part-of-Speech-Tagging (PoS-Tagging)



NLP = Natural Language Processing

Exemplary NLP areas and tasks:

- Tokenization
- Lemmatization
- Part-of-Speech-Tagging (PoS-Tagging)
- Sentiment Analysis



Sentiment Analysis

NLP = Natural Language Processing

Exemplary NLP areas and tasks:

- Tokenization
- Lemmatization
- Part-of-Speech-Tagging (PoS-Tagging)
- Sentiment Analysis
- Named Entity Recognition (NER)

On the 15th of September DATE , Tim Cook PERSON announced that
Apple ORG wants to acquire ABC Group ORG from New York GPE
for 1 billion dollars MONEY

Natural Language Toolkit (NLTK)

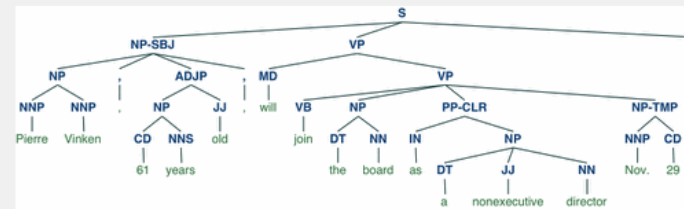
- Python library for NLP
- <https://www.nltk.org/>

Tokenize and tag some text:

```
>>> import nltk
>>> sentence = """At eight o'clock on Thursday morning
... Arthur didn't feel very good."""
>>> tokens = nltk.word_tokenize(sentence)
>>> tokens
['At', 'eight', "o'clock", 'on', 'Thursday', 'morning',
 'Arthur', 'did', "n't", 'feel', 'very', 'good', '.']
>>> tagged = nltk.pos_tag(tokens)
>>> tagged[0:6]
[('At', 'IN'), ('eight', 'CD'), ('o'clock', 'JJ'), ('on', 'IN'),
 ('Thursday', 'NNP'), ('morning', 'NN')]
```

Display a parse tree:

```
>>> from nltk.corpus import treebank
>>> t = treebank.parsed_sents('wsj_0001.mrg')[0]
>>> t.draw()
```



Identify named entities:

```
>>> entities = nltk.chunk.ne_chunk(tagged)
>>> entities
Tree('S', [(('At', 'IN'), ('eight', 'CD'), ('o'clock', 'JJ'),
            ('on', 'IN'), ('Thursday', 'NNP'), ('morning', 'NN')),
            Tree('PERSON', [(('Arthur', 'NNP')]),
            ('did', 'VBD'), ("n't", 'RB'), ('feel', 'VB'),
            ('very', 'RB'), ('good', 'JJ'), ('.', '.')]])
```

spaCy

<https://spacy.io/>

Python library for Natural Language Processing
(Tokenization, Pos-Tagging, NER, Sentiment Analysis, ...)



```
# pip install -U spacy
# python -m spacy download en_core_web_sm
import spacy
```

```
# Load English tokenizer, tagger, parser and NER
nlp = spacy.load("en_core_web_sm")
```

```
# Process whole documents
```

```
text = ("When Sebastian Thrun started working on self-driving cars at "
        "Google in 2007, few people outside of the company took him "
        "seriously. "I can tell you very senior CEOs of major American "
        "car companies would shake my hand and turn away because I wasn't "
        "worth talking to," said Thrun, in an interview with Recode earlier "
        "this week.")
```

```
doc = nlp(text)
```

```
# Analyze syntax
```

```
print("Noun phrases:", [chunk.text for chunk in doc.noun_chunks])
print("Verbs:", [token.lemma_ for token in doc if token.pos_ == "VERB"])
```

```
# Find named entities, phrases and concepts
```

```
for entity in doc.ents:
    print(entity.text, entity.label_)
```

RUN

Industrial-Strength Natural Language Processing

IN PYTHON



Hugging Face



Models



Datasets



Spaces

<https://huggingface.co/>

NEW Create Assistants in HuggingChat



The AI community building the future.

The platform where the machine learning community
collaborates on models, datasets, and applications.

Tasks Libraries Datasets Languages Licenses Other

Q Filter Tasks by name

Multimodal

Text-to-Image Image-to-Text
Text-to-Video Visual Question Answering
Document Question Answering Graph Machine Learning

Computer Vision

Depth Estimation Image Classification
Object Detection Image Segmentation
Image-to-Image Unconditional Image Generation
Video Classification Zero-Shot Image Classification

Natural Language Processing

Text Classification Token Classification
Table Question Answering Question Answering
Zero-Shot Classification Translation
Summarization Conversational
Text Generation Text2Text Generation
Sentence Similarity

Audio

Text-to-Speech Automatic Speech Recognition
Audio-to-Audio Audio Classification
Voice Activity Detection

Models 469,541 Filter by name

meta-llama/Llama-2-70b
Text Generation • Updated 4 days ago • 25.2k • 64

stabilityai/stable-diffusion-xl-base-0.9
Updated 6 days ago • 2.01k • 393

openchat/openchat
Text Generation • Updated 2 days ago • 1.3k • 136

llyasviel/ControlNet-v1-1
Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL
Updated 3 days ago • 2.66k • 334

meta-llama/Llama-2-13b
Text Generation • Updated 4 days ago • 328 • 64

tiiuae/falcon-40b-instruct
Text Generation • Updated 27 days ago • 288k • 899

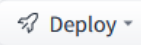
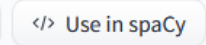
WizardLM/WizardCoder-15B-V1.0
Text Generation • Updated 3 days ago • 12.5k • 332

CompVis/stable-diffusion-v1-4
Text-to-Image • Updated about 17 hours ago • 448k • 5.72k

 **easyh/de_fnhd_ne1dh**   like 0

 Token Classification  spaCy  German  Eval Results

 **Model card**  Files and versions  Community

  Deploy 

 Edit model card

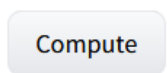
Deutsche NER-Pipeline für frühneuhochdeutsche Texte (2.Version)

Feature	Description
Name	de_fnhd_ne1dh
Version	0.0.2
spaCy	>=3.4.1,<3.5.0
Default Pipeline	tok2vec, ner
Components	tok2vec, ner
Vectors	500000 keys, 500000 unique vectors (300 dimensions)
Sources	n/a
License	n/a

Downloads last month
0

 **Hosted inference API** 

 Token Classification



This model can be loaded on the Inference API on-demand.

 JSON Output 

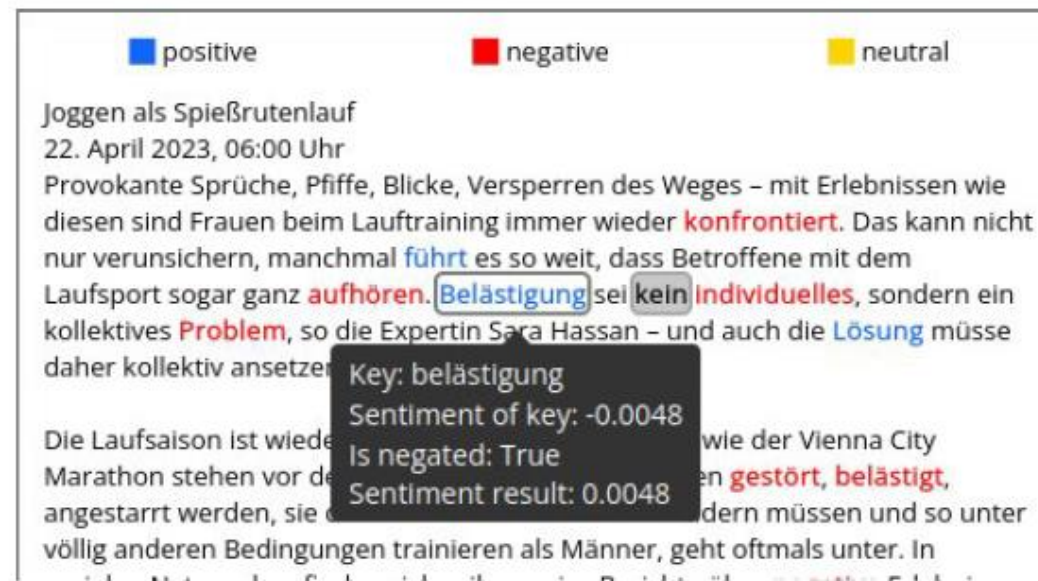
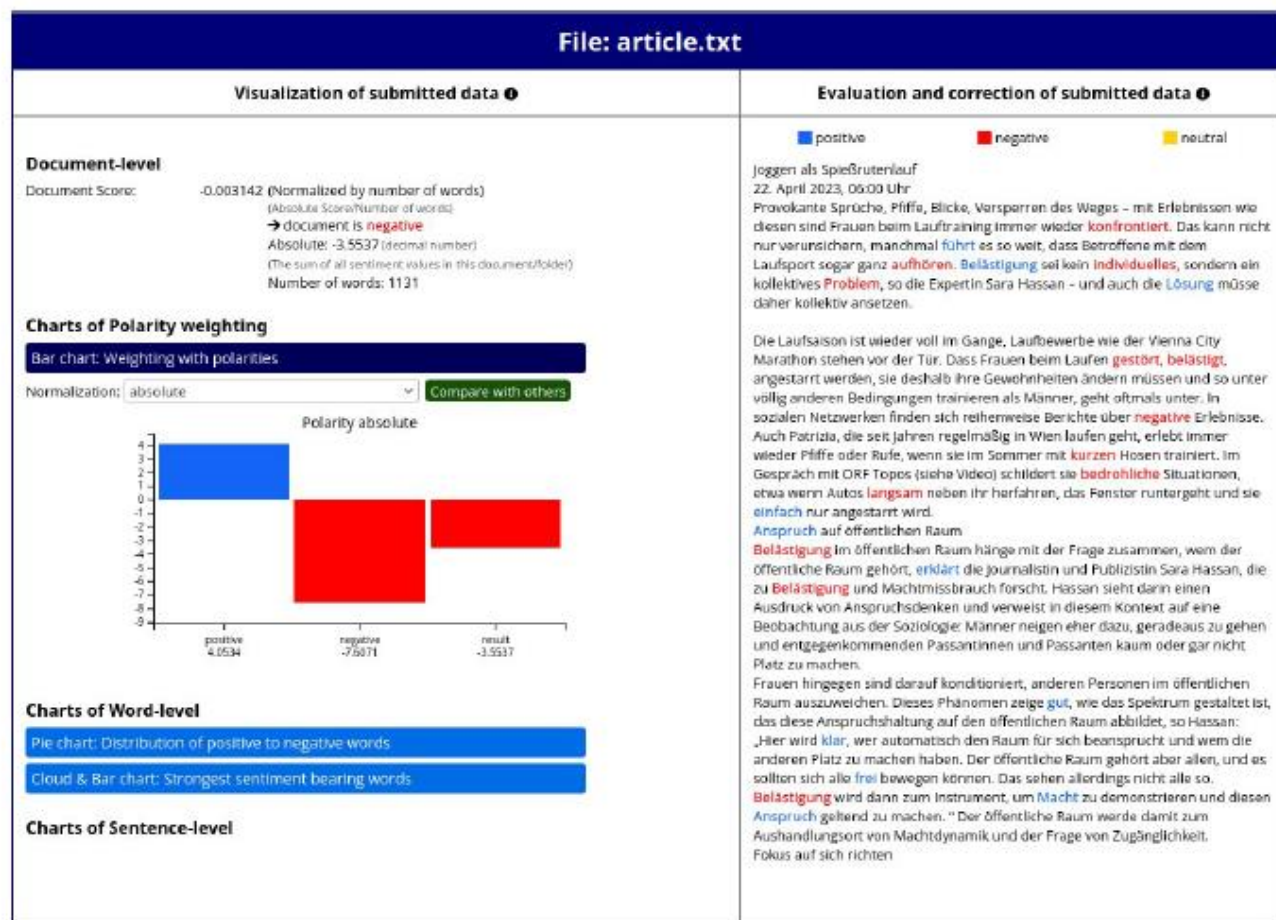
 **Space using easyh/de_fnhd_ne1dh** 1

  easyh/NerDH_Visualisierer

 **Evaluation results** 

SentText

<https://thomasschmidtur.pythonanywhere.com/>



Tutorials and further resources

- Tutorial for sentiment analysis:

<https://programminghistorian.org/en/lessons/sentiment-analysis>

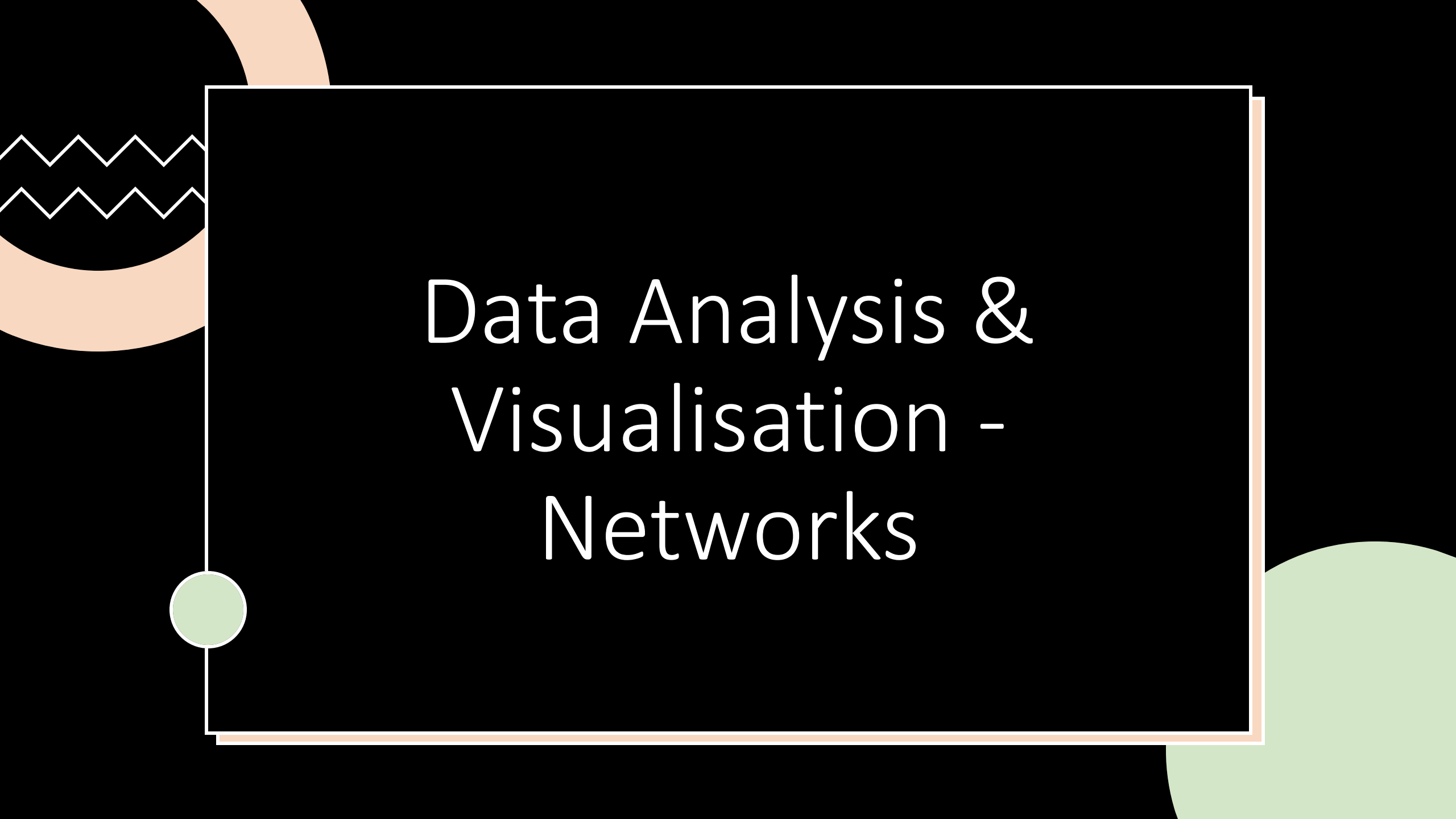
- Tutorial for spacy:

<https://programminghistorian.org/en/lessons/corpus-analysis-with-spacy>

- Tutorial for SentText: <https://fortext.net/tools/tools/senttext>

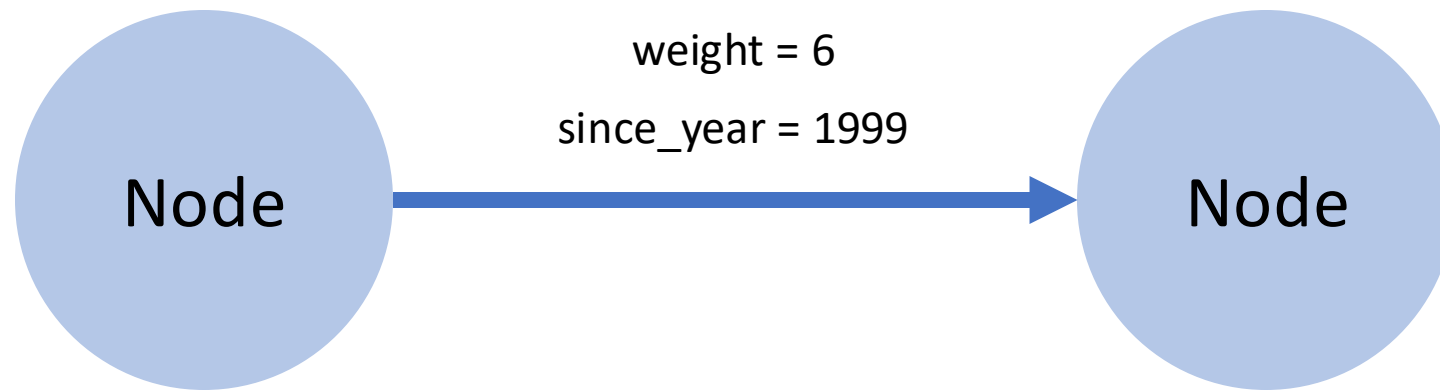
- Tutorial for NLTK: <https://www.nltk.org/book/ch01.html>

- Collection of tools for corpus analysis: <https://corpus-analysis.com/>



Data Analysis & Visualisation - Networks

Nodes and edges



Graph

Ezlinavis

<https://ezlinavis.dracor.org>

Easy Linavis

Examples ▾

Graph ▾

About

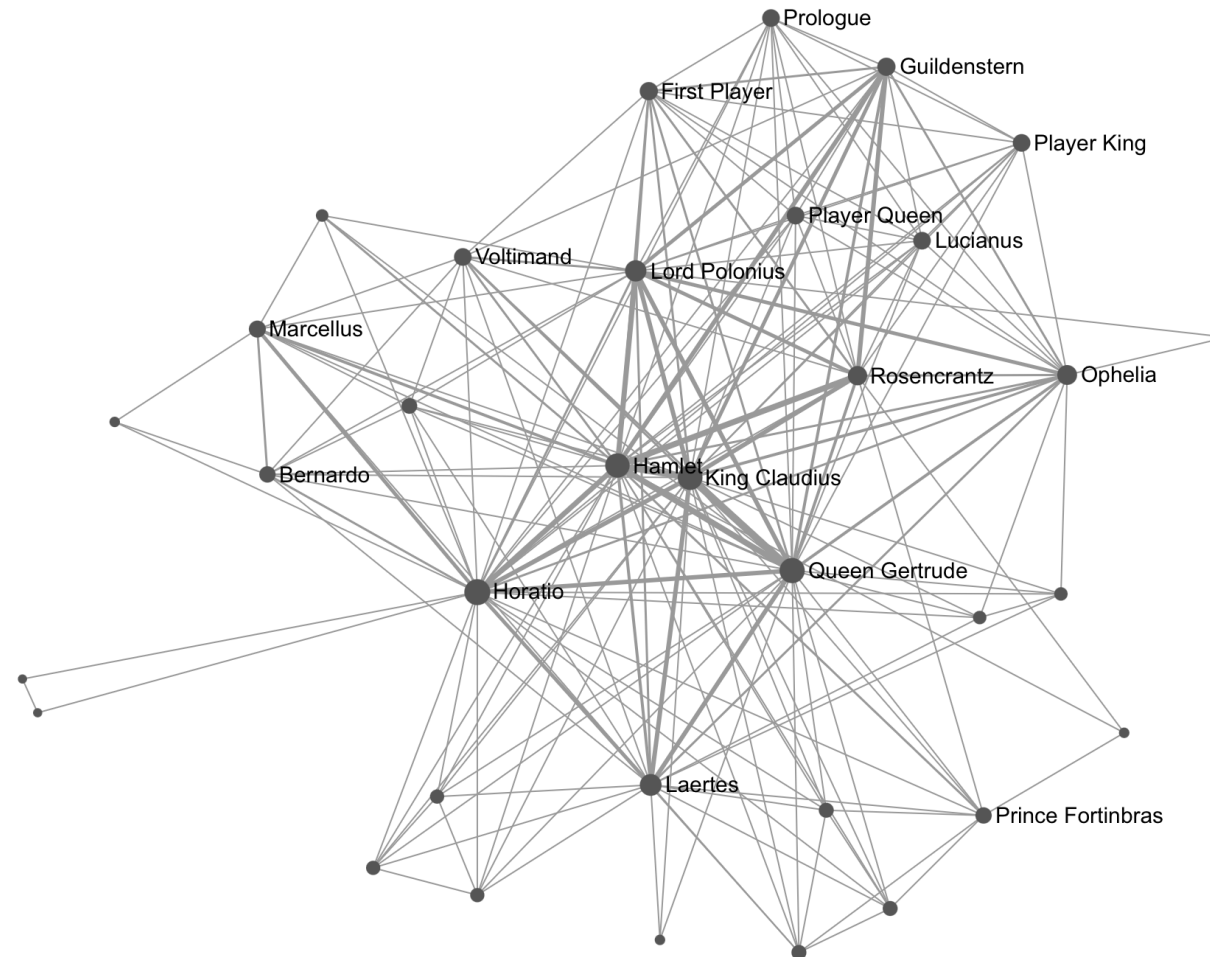
Shakespeare: The Tragedy of Hamlet,
Prince of Denmark (c. 1599/1602)
Characters/speakers per scene, in
order of appearance
Extracted from this version:
<http://shakespeare.mit.edu/hamlet/full.html>
By Frank Fischer, October 2017
Formalisation starts with first hashtag:

```
# ACT I
## SCENE I. Elsinore. A platform before
the castle.
Bernardo
Francisco
Horatio
Marcellus
## SCENE II. A room of state in the
castle.
King Claudius
Cornelius
Voltimand
Laertes
Lord Polonius
Hamlet
Queen Gertrude
Horatio
Marcellus
Bernardo
## SCENE III. A room in Polonius'
house.
Laertes
Ophelia
Lord Polonius
## SCENE IV. The platform.
Hamlet
Horatio
Marcellus
## SCENE V. Another part of the
platform.
Hamlet
Ghost
Marcellus
Horatio
# ACT II
```

[download CSV](#)

Source,Type,Target,Weight

```
Bernardo,Undirected,Cornelius,1
Bernardo,Undirected,Francisco,1
Bernardo,Undirected,Hamlet,1
Bernardo,Undirected,Horatio,2
Bernardo,Undirected,King Claudius,1
Bernardo,Undirected,Laertes,1
Bernardo,Undirected,Lord Polonius,1
Bernardo,Undirected,Marcellus,2
Bernardo,Undirected,Queen Gertrude,1
Bernardo,Undirected,Voltimand,1
Captain,Undirected,Hamlet,1
Captain,Undirected,Prince Fortinbras,1
Captain,Undirected,Rosencrantz,1
Cornelius,Undirected,Hamlet,1
Cornelius,Undirected,Horatio,1
Cornelius,Undirected,King Claudius,1
Cornelius,Undirected,Laertes,1
Cornelius,Undirected,Lord Polonius,1
Cornelius,Undirected,Marcellus,1
Cornelius,Undirected,Queen Gertrude,1
Cornelius,Undirected,Voltimand,1
Danes,Undirected,Gentleman,1
Danes,Undirected,Horatio,1
Danes,Undirected,King Claudius,1
Danes,Undirected,Laertes,1
Danes,Undirected,Ophelia,1
Danes,Undirected,Queen Gertrude,1
First Ambassador,Undirected,Hamlet,1
First Ambassador,Undirected,Horatio,1
First Ambassador,Undirected,King Claudius,1
First Ambassador,Undirected,Laertes,1
First Ambassador,Undirected,Lord Polonius,1
```



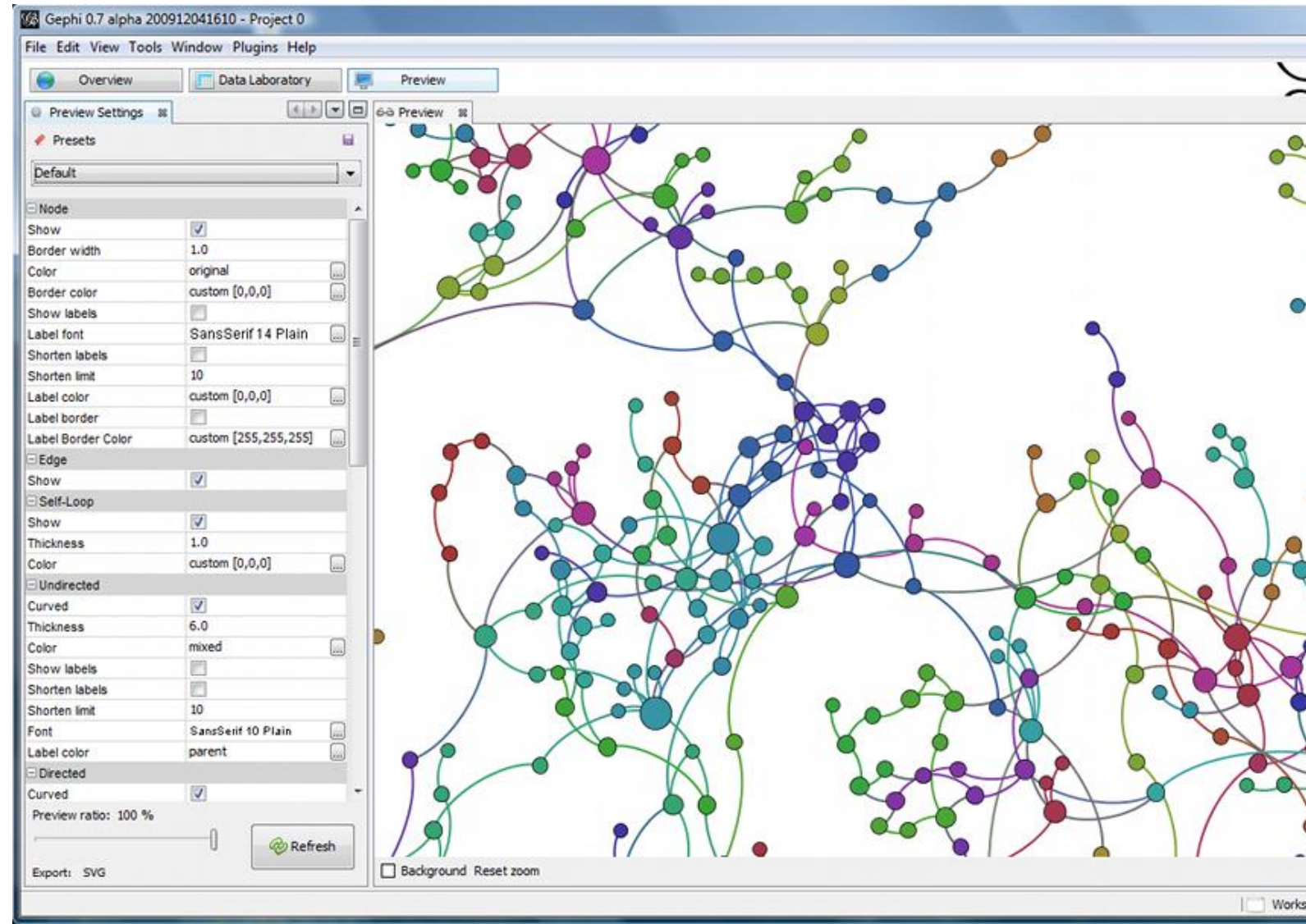
Gephi



Textual and spatial encoding

- ❖ LOD
- ❖ Relational data/graphs
- ❖ Various downloadable formats

Online version: <https://gephi.org/gephi-lite/>





NetworkX

Network Analysis in Python

[Documentation](#)

- Create **graphs** from scratch or from pre-existing data
- Run **analyses** on them, e.g.:
 - How many connections?
 - How strong are the connections?
- Basic **visualisation** methods

```
nx.degree(G)
```

```
[8]
```

```
... DegreeView({'Aemon': 5, 'Grenn': 4, 'Samwell': 15, 'Aerys': 4, 'Jaime': 24, 'Robert':
```

```
degrees = dict(nx.degree(G))
nx.set_node_attributes(G, name='degree', values=degrees)
```

```
[9]
```

```
degree_df = pd.DataFrame(G.nodes(data='degree'), columns=['node', 'degree'])
degree_df = degree_df.sort_values(by='degree', ascending=False)
degree_df.head()
```

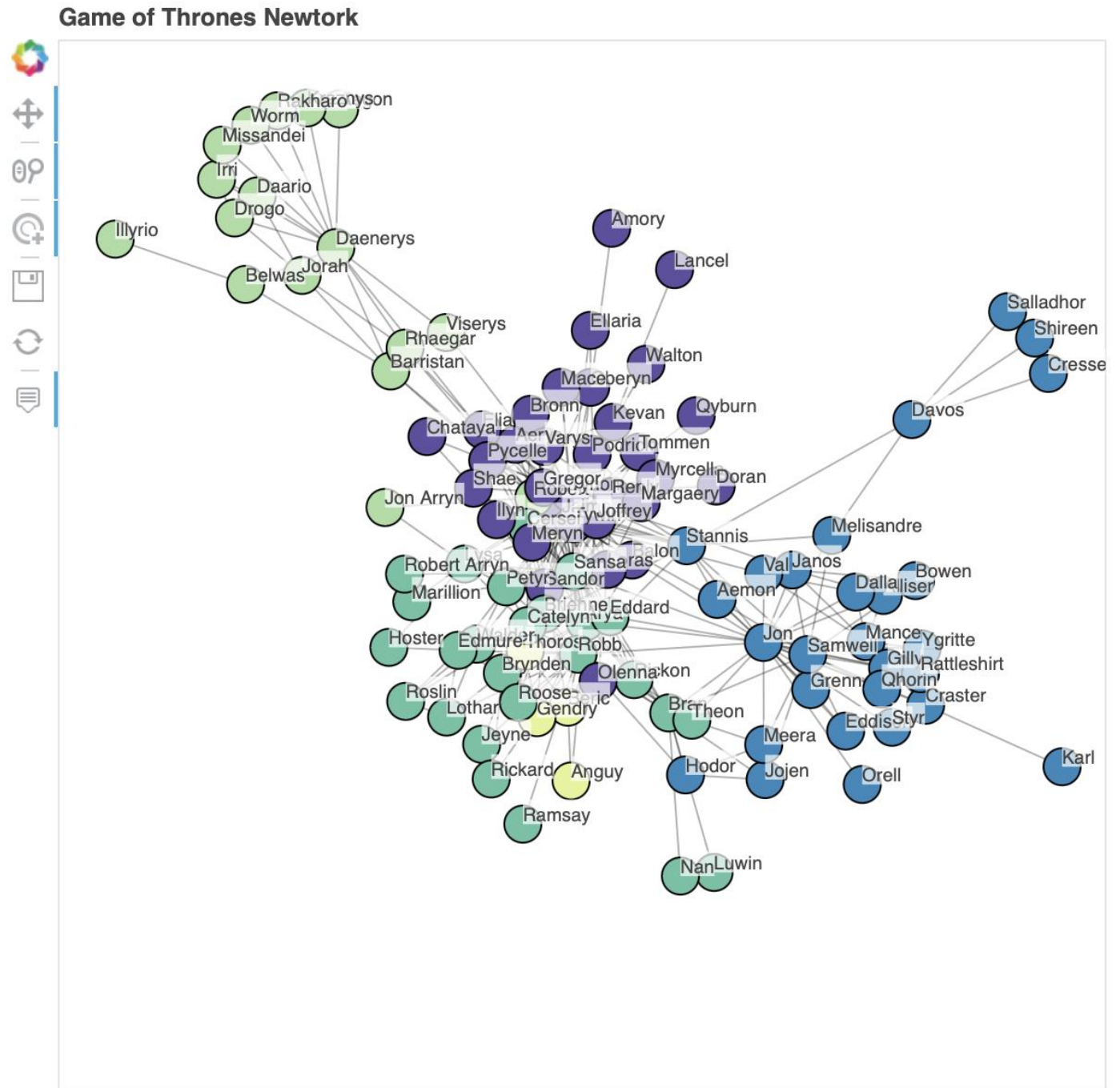
```
[13]
```

```
...
   node degree
6  Tyrion    36
21   Jon    26
50  Sansa    26
48   Robb    25
4   Jaime    24
```



<http://bokeh.org>

- **Python** library
- Create **interactive** visualisations
- Use these visualisations in **web pages**
- See an example [here](#) (based on *Game of Thrones*)

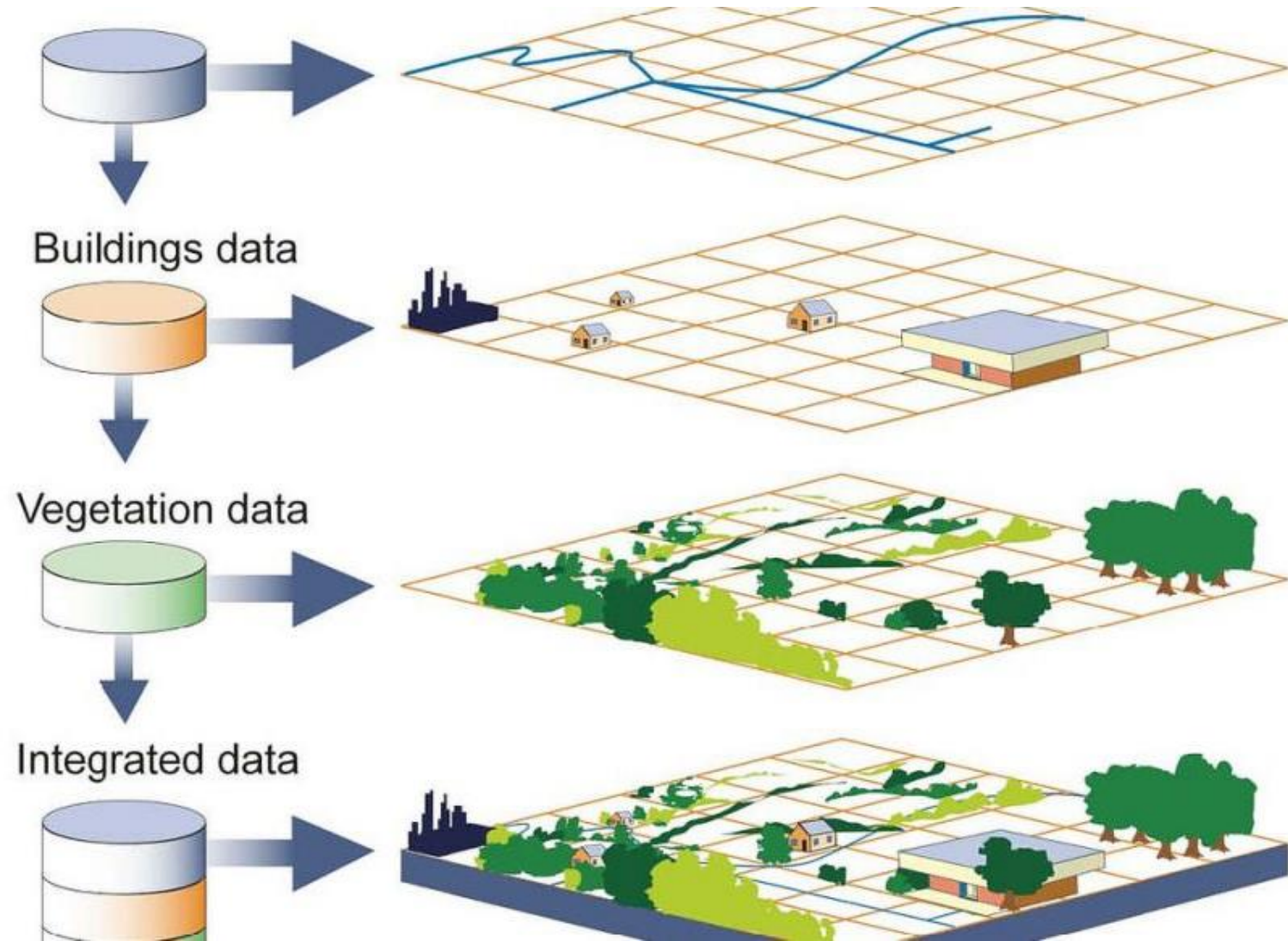


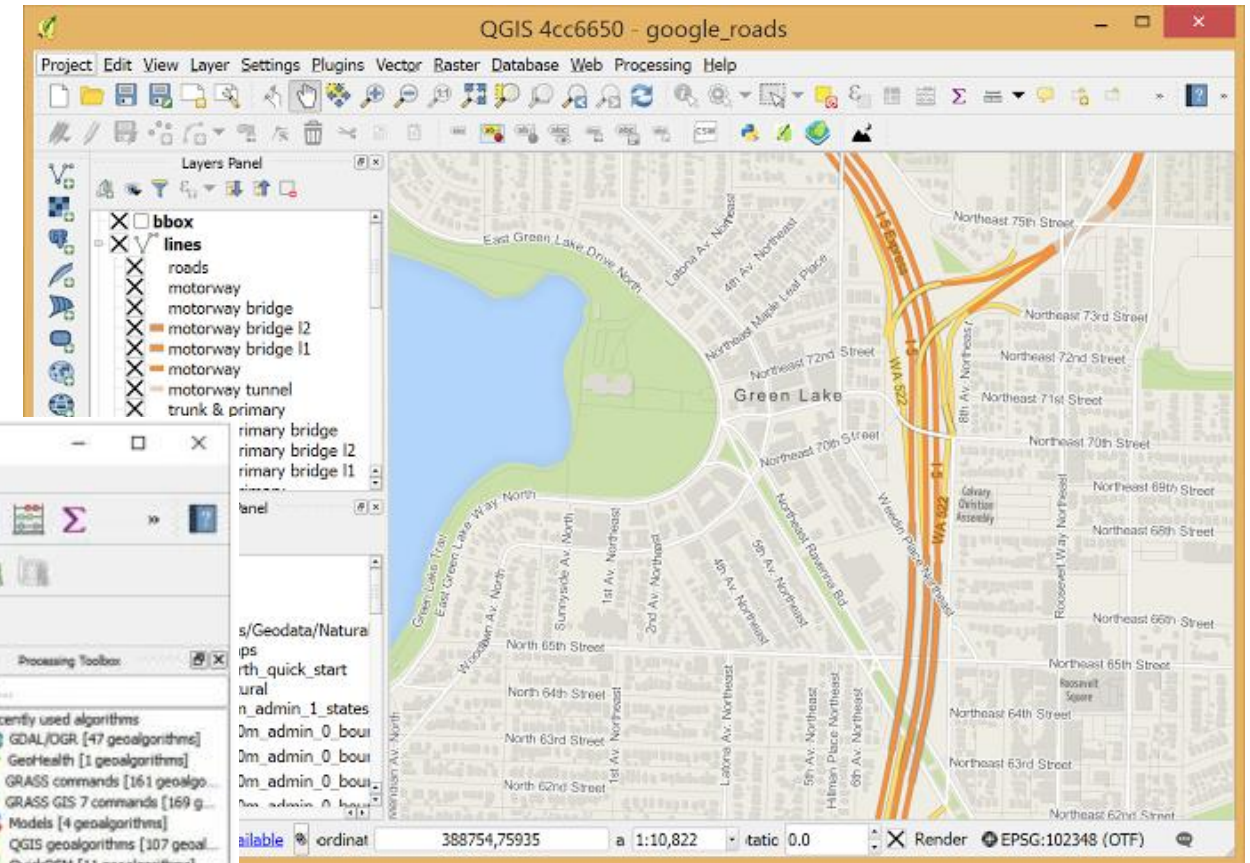
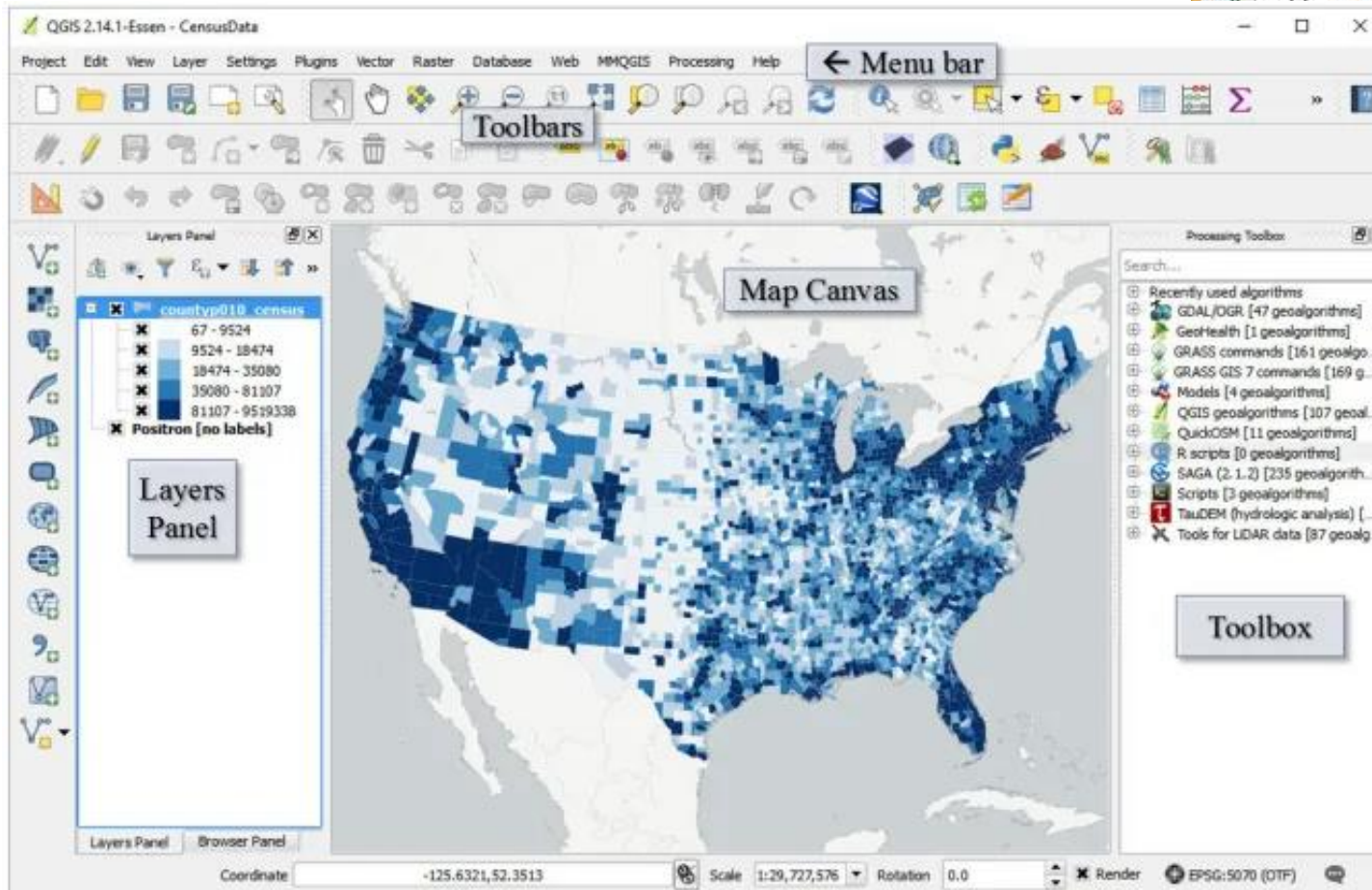


GIS

GIS = Geographic Information System(s)

- computer-based tools used to store, visualize, analyze, and interpret geographic data





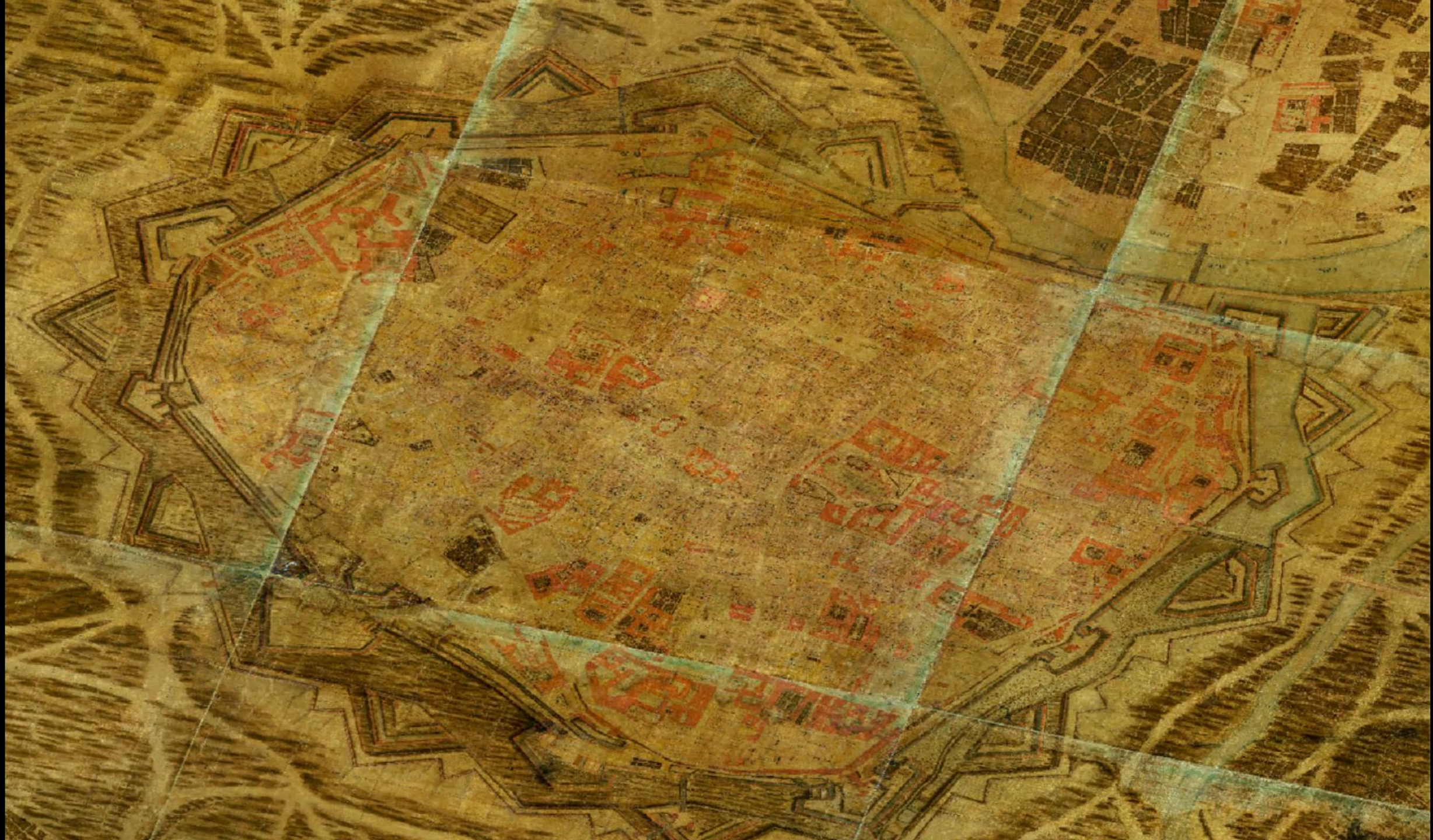
<https://qgis.org/de/site/>

Open-Source GIS software
(spatial analysis, map
visualisations, ...)

Exemplary Use Case

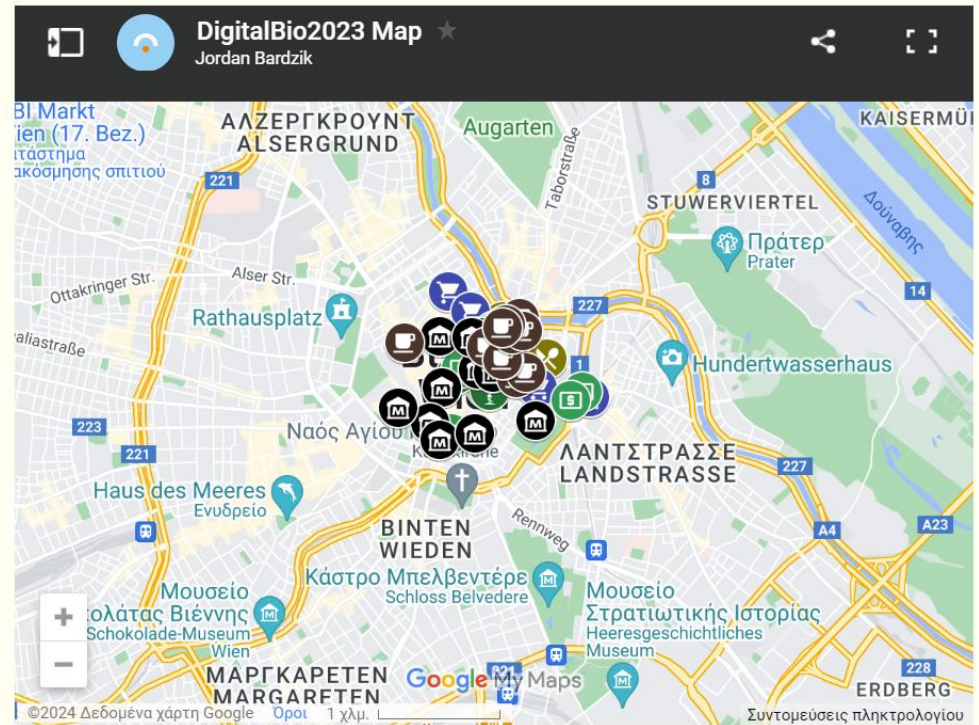


- ACDH-CH Project "Visiting Vienna"
- Mapping of arriving persons in early 18th century Vienna based on arrival list data
- Use of georeferenced historic plans of Vienna



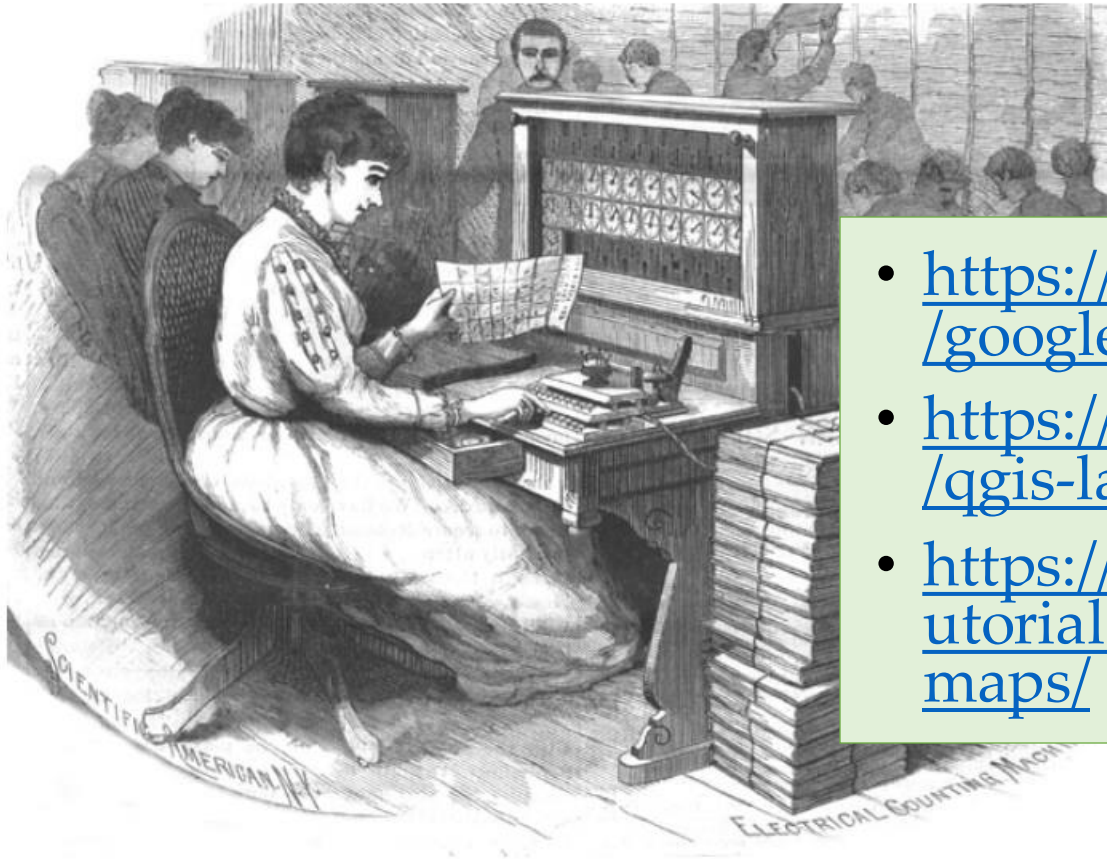


pharmacies, restaurants, cafes, ATMs, and signs with Google Maps.

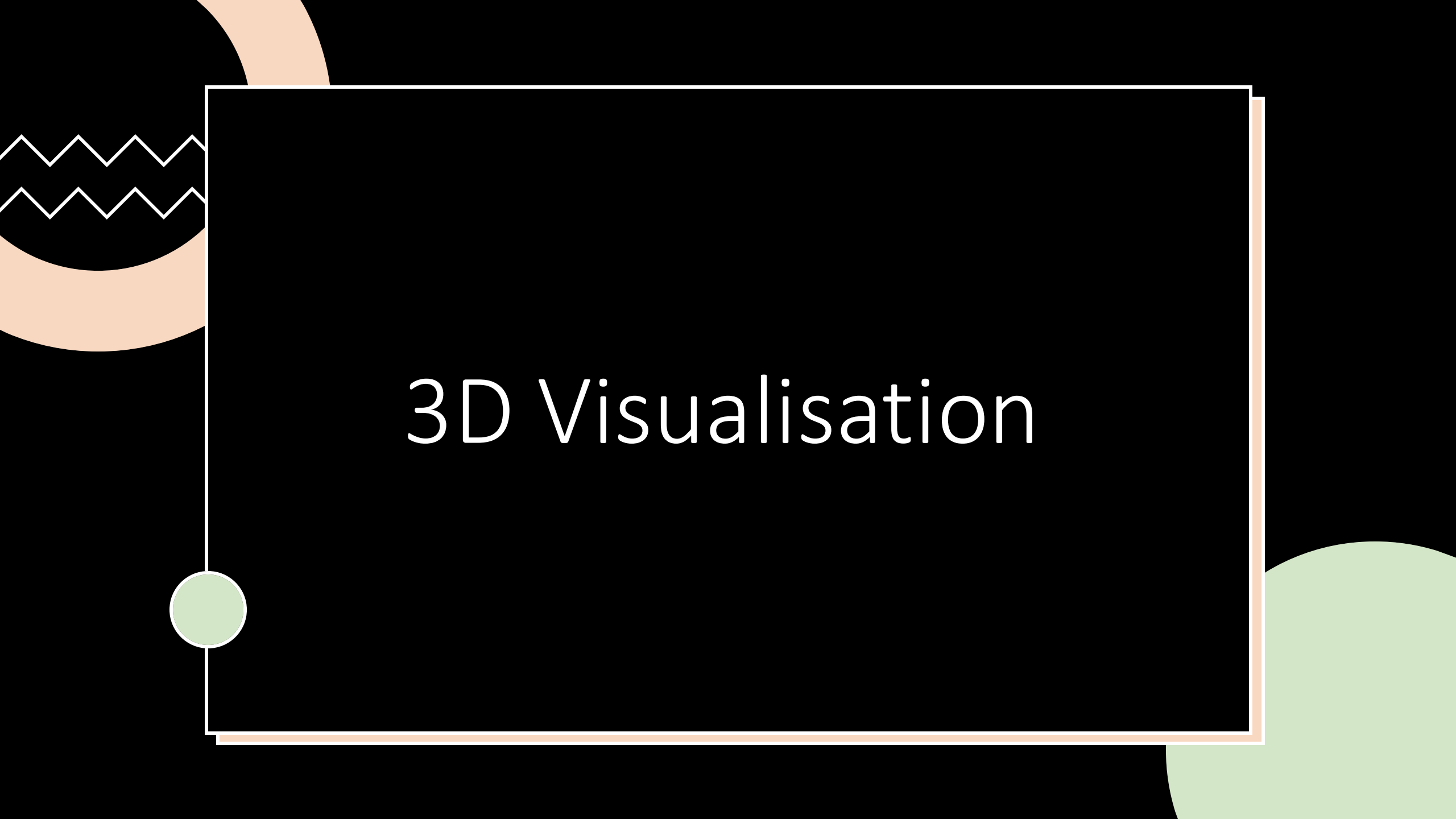


https://www.google.com/maps/d/edit?mid=1o1p_2DPFs62j6OXx00ShFAvqg1JCIQw&usp=sharing

Tutorials



- <https://programminghistorian.org/en/lessons/googlemaps-googleearth>
- <https://programminghistorian.org/en/lessons/qgis-layers>
- <https://multimedia.journalism.berkeley.edu/tutorials/create-custom-maps-in-google-maps/>

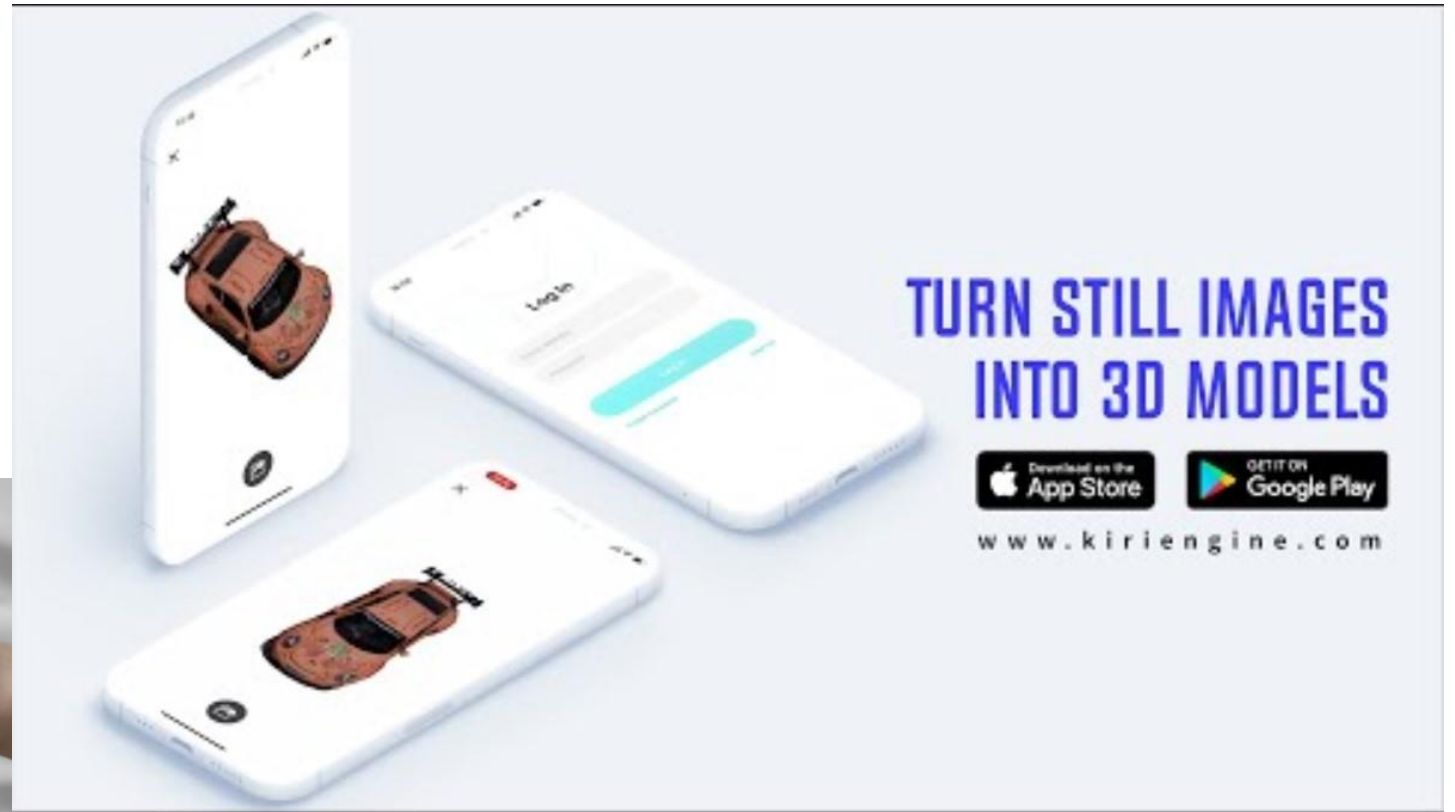


3D Visualisation



KIRI Engine

<https://www.kiriengine.app/>

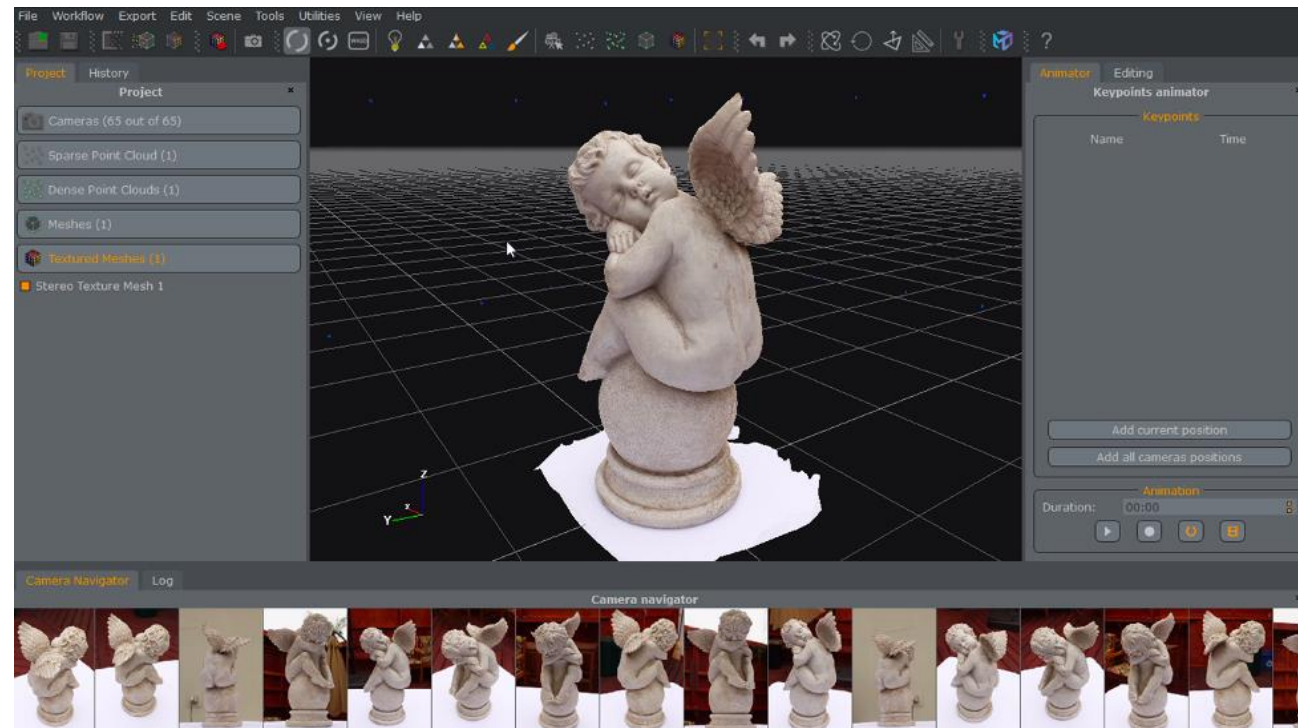
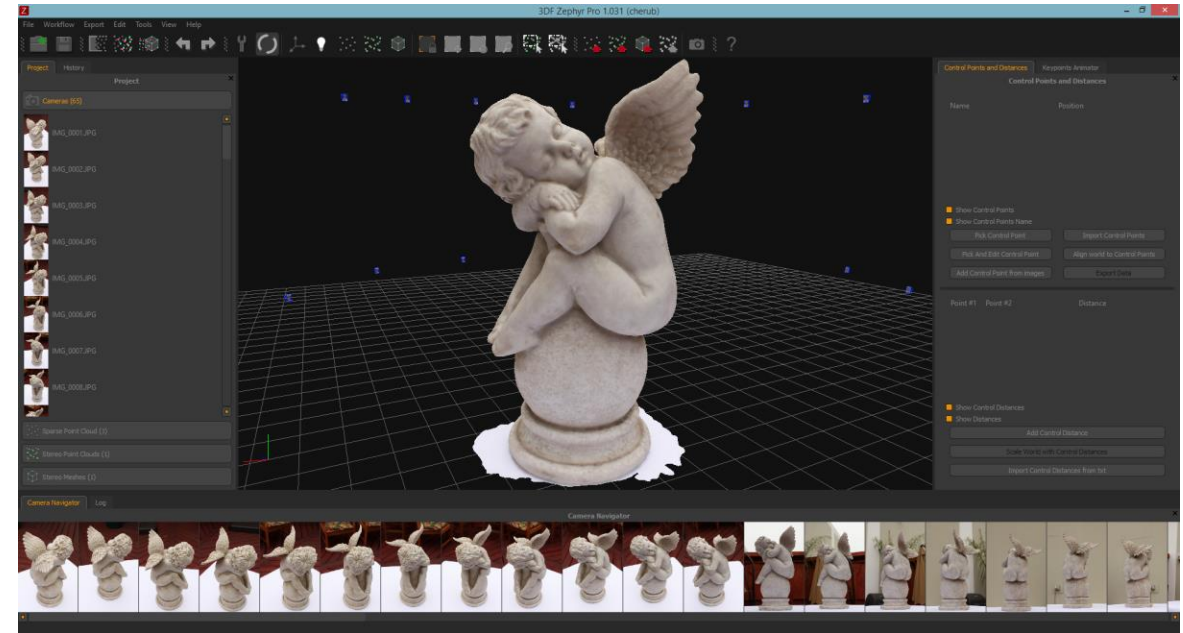




3DF ZEPHYR

The Complete Photogrammetry Solution

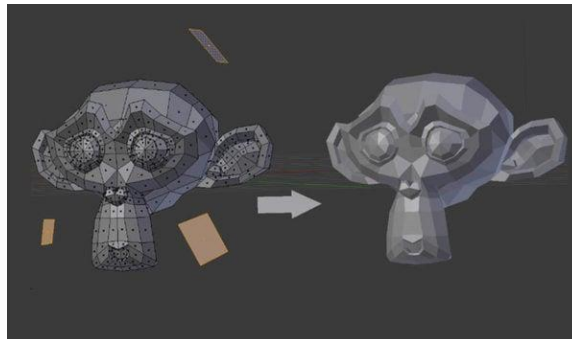
<https://www.3dflow.net/3df-zephyr-photogrammetry-software/>



Free Photogrammetry tool (3D representations)



<https://www.blender.org/>



Free and Open source 3D creation suite.

It supports the entirety of the 3D pipeline (e.g. modeling, rigging, animation, simulation, rendering, compositing and motion tracking, even video editing and game creation.)



Semantic Web



WIKIPEDIA
The Free Encyclopedia

<https://en.wikipedia.org/wiki/Vienna>



WIKIPEDIA
The Free Encyclopedia

Q Create account Log in ...

Vienna

226 languages

Article Talk

Read View source View history Tools

From Wikipedia, the free encyclopedia

Coordinates: 48°12′30″N 16°22′21″E﻿ / ﻿48.20833°N 16.37250°E﻿ / 48.20833; 16.37250

This article is about the capital of Austria. For other uses, see [Vienna \(disambiguation\)](#).

"Wien" redirects here. For other uses, see [Wien \(disambiguation\)](#).

Not to be confused with [Vienne \(disambiguation\)](#).



This article **contains weasel words: vague phrasing that often accompanies biased or unverifiable information.** Such statements should be **clarified or removed.** *(September 2023)*

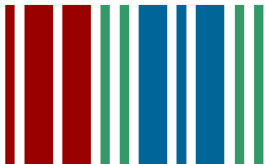
Vienna (/ˈviːnə/ [ⓘ] [ⓘ] *vee-EN-ə*; [ⓘ] German: *Wien* [ⓘ] [ⓘ]; [ⓘ] [ⓘ] *Austro-Bavarian: *Wean** [ⓘ] [ⓘ]) is the capital, largest city, and one of **nine federal states** of **Austria**. Vienna is Austria's **most populous city** and its **primate city**, with about two million inhabitants^{[3][11]} (2.9 million within the **metropolitan area**,^[12] nearly one-third of the country's population), and its **cultural**, **economic**, and **political** center. It is the **fifth-largest** city proper by population in the **European Union** and the largest of all **cities on the Danube river** by population.

The city lies on the eastern edge of the **Vienna Woods** (*Wienerwald*)—the northeasternmost foothills of the **Alps**, that separate Vienna from the more western parts of Austria—at

Vienna

Wien (**German**)
Wean (**Bavarian**)

Capital city, federal state and municipality



WIKIDATA

<https://www.wikidata.org/wiki/Q1741>

Vienna (Q1741)

capital of and state in Austria

Wien | Vienna, Austria | W

In more languages

Configure

Language	Label	Description	Also known as
English	Vienna	capital of and state in Austria	Wien Vienna, Austria W
German	Wien	Bundeshauptstadt der Republik Österreich	AT-9 AT VIE W
Bavarian	Wean	Hauptstod vo Österreich	
French	Vienne	capitale de l'Autriche	Wien

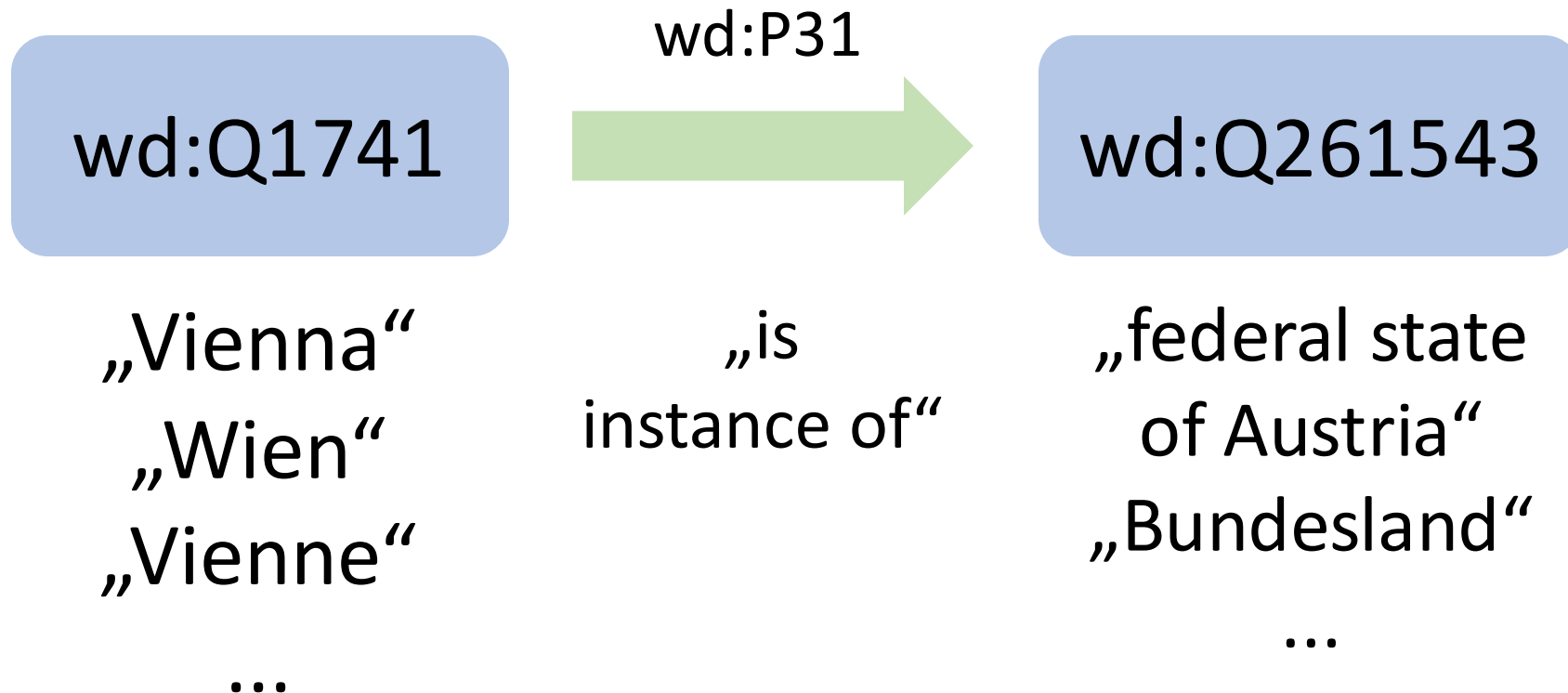
All entered languages

Statements

instance of	 federal capital 0 references
	 statutory city of Austria 0 references
	 federal state of Austria start time 18 November 1920 <i>Gregorian</i>

How does the Semantic Web work?

Vienna (/ˈviːɛnə/ ⓘ) *vee-EN-ə*; ^[9]^[10] German: *Wien* [ˈviːn] ⓘ); **Austro-Bavarian**: *Wean* [ˈvɛŋ]) is the capital, largest city, and one of nine federal states of Austria. Vienna is Austria's most populous city and its primate city,



Google Knowledge Panel



harrison ford



M

All

Images

News

Videos

Books

More

Tools

SafeSearch

About 81.500.000 results (0,40 seconds)

Harrison Ford

American actor

Overview

Movies

Relationships

Videos



Ethan Miller/FilmMagic



Screen Rant

Jake Gyllenhaal's Upcoming Remake Of \$221 Million ...

... Harrison Ford film looks like it could be even better. Jake Gyllenhaal in Road...

3 days ago

Age

81 years
13 Jul 1942

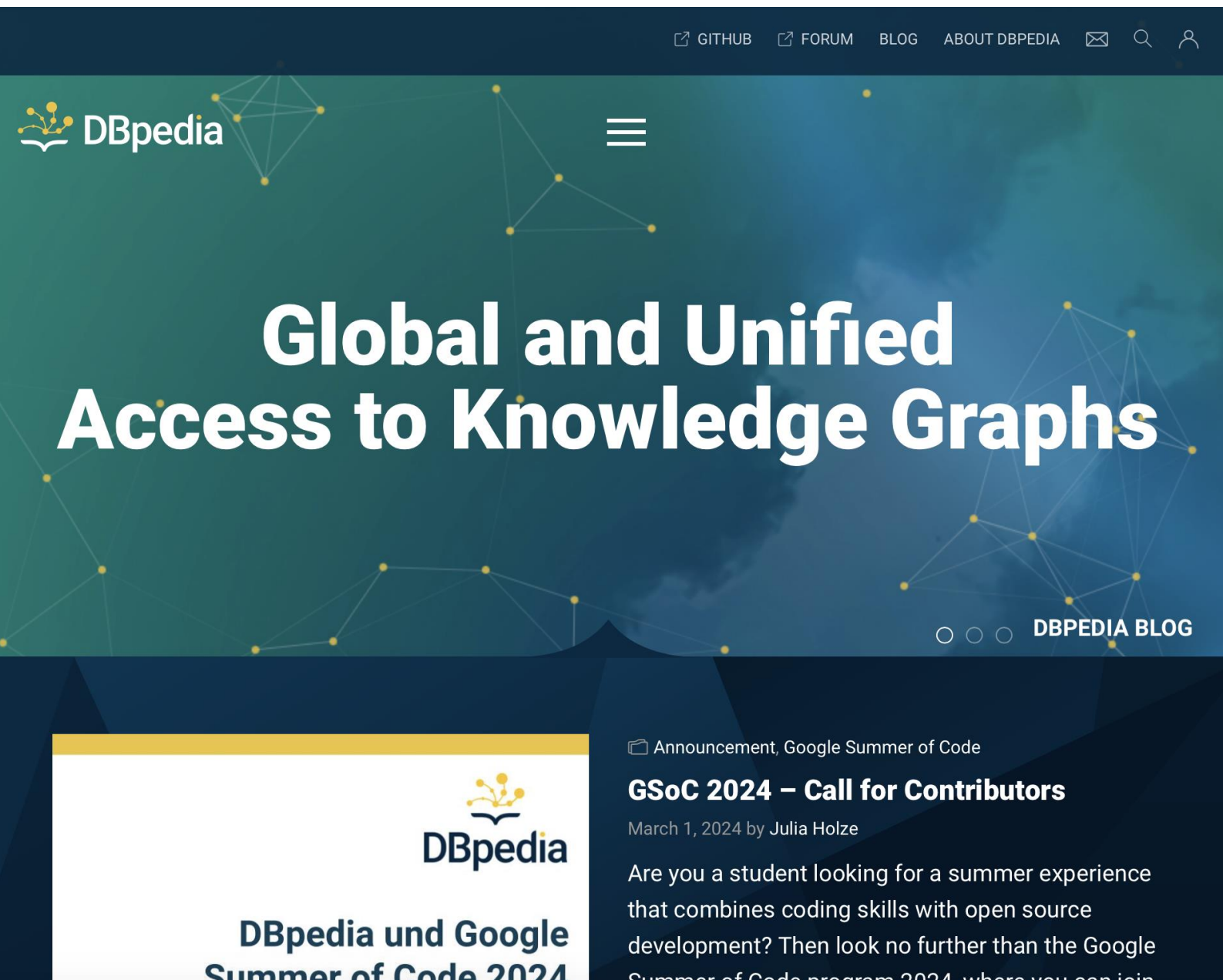
Spouse

Calista Flockhart (m. 2010), Melissa Mathison (m....



7:27

10 Jul 2023



[Faceted browser](#)

[Query interface](#)
(advanced)

<https://dbpedia.org/page/Vienna>

Controlled vocabularies



Vocabularies About Feedback Help | Interface language: English ▾

UNESCO Thesaurus

Content language English ▾ x Search

Alphabetical

Hierarchy

Groups

A Å B C D E F G H I J K
L M N O P Q R S T U V W
X Y Z

Abadi → *Awadhi*
Abandoned children
Abbreviations
Ability
Ability grouping
Abohi → *Awadhi*
Aboriginals → *Indigenous peoples*
Abortion
Absenteeism → *Leave*
Abstract journals → *Abstracts*
Abstract reasoning → *Reasoning*
Abstracting
Abstracting and indexing services → *Bibliographic services*
Abstracts
Abuse of human rights → *Human rights violations*
Abuse of power → *Oppression*
Academic achievement
Academic admission → *Admission requirements*
Academic buildings
Academic degrees → *Degrees*
Academic equipment → *Educational equipment*
Academic facilities → *Educational facilities*
Academic failure
Academic fraud
Academic freedom
Academic grouping → *Educational grouping*
Academic laboratories → *University laboratories*
Academic libraries

Vocabulary information

TITLE	UNESCO Thesaurus
DESCRIPTION	The UNESCO Thesaurus is a controlled and structured list of terms used in subject analysis and retrieval of documents and publications in the fields of education, culture, natural sciences, social and human sciences, communication and information. Continuously enriched and updated, its multidisciplinary terminology reflects the evolution of UNESCO's programmes and activities.
IDENTIFIER	http://vocabularies.unesco.org/thesaurus
PUBLISHER	UNESCO
RIGHTS HOLDER	UNESCO
RIGHTS	CC-BY-SA
LICENSE	http://creativecommons.org/licenses/by-sa/3.0/igo/
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- Define **terms** that should be used in a domain / for a task
- Allow all applications (or users) to understand what we are talking about **unequivocally!**
- Usually based on the [SKOS](#) model
- Some examples include the [UNESCO Thesaurus](#) (on the left)

Some ways to go...

- Search for any Wikipedia page that might interest you, and click on „**Tools > Wikidata item**“... you can compare the structured data to the original page or browse through the entities by clicking on the links
- (More advanced) You can load a sample Neo4j graph database by using the [Neo4j Sandbox](#), for example a database on „Movies“
- Have a look at the ACDH-CH services, [ARCHE](#) and [Vocabs](#), which provide data according to Semantic Web recommendations

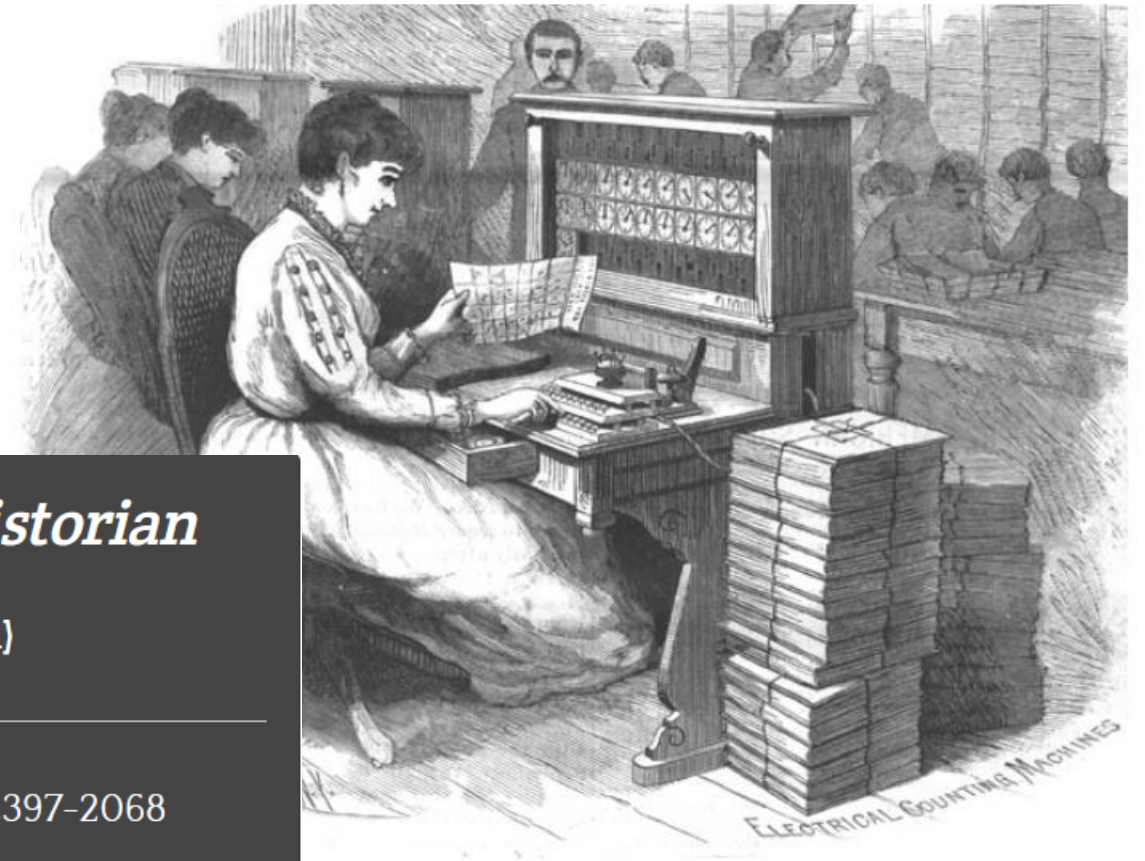


Further resources to educate yourself



Programming Historian

<https://programminghistorian.org/>



Enter *Programming Historian*

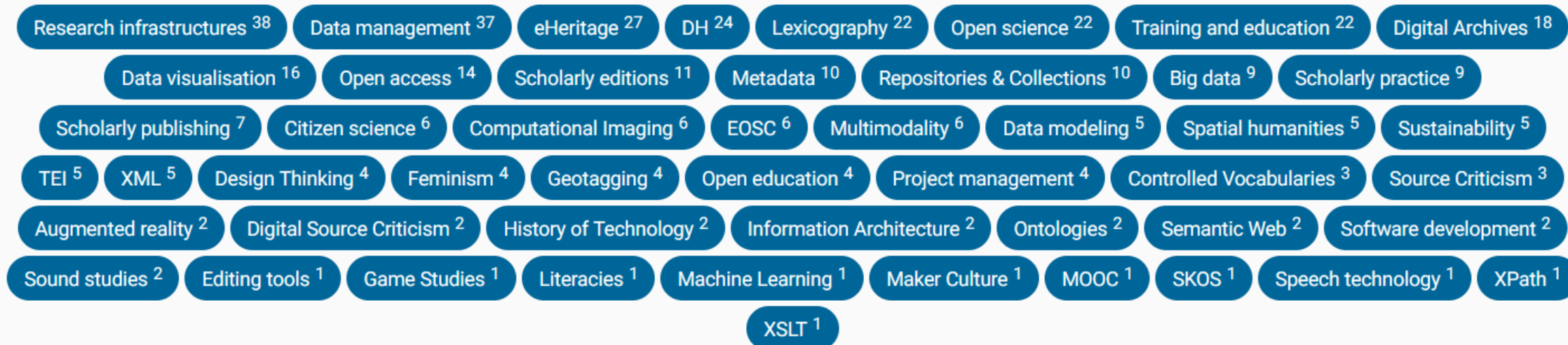
(The initial English version)

[PH]

101 lessons

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We publish novice-friendly, peer-reviewed tutorials that help humanists learn a wide range of digital tools, techniques, and workflows to facilitate research and teaching.



Looking for learning resources?

DARIAH-CAMPUS is a discovery framework and hosting platform for learning resources.

ACDH-CH-HowTo

<https://howto.acdh.oeaw.ac.at/>

Share and expand your knowledge in Digital Humanities

Welcome to ACDH-CH Learning Resources. This website gathers interactive learning material, practical HowTo articles and best practices on a wide range of Digital Humanities topics, methodologies and infrastructures. These resources can be used for self-guided learning as well as for teaching in higher education. It is part of our mission to actively transfer knowledge from ongoing research into the wider Digital Humanities Community as well as to educate a new generation of humanities researchers with digital methods.

Posts



Corpus Query Language im Austrian Media Corpus

Diese Resource stellt den Austria Media Corpus (amc) und seine Nutzungsmöglichkeiten vor. Im Besonderen wird die Corpus Query Language (CQL) erklärt.

[Read more →](#)



Lesson 1: Make a plan, then don't stick to it

Fine-tuning a workflow and a question millions of years old. This is part of a blog post series documenting the SemanticKraus research project at the ACDH-CH.

[Read more →](#)



Lesson 2: Meet the data. Examples of data perspectivity, with some thoughts on shopping for groceries.

This is part of a blog post series documenting the SemanticKraus research project at the ACDH-CH.

[Read more →](#)



How to do a HowTo - A short manual for the ACDH-CH Learning Resources Platform

This article is a short introduction to the HowTo-Platform of the ACDH-CH and covers how to plan and structure content, write an article and publish it via the Content Management System.

[Read more →](#)



ACDH-CH TOOL GALLERIES



TRANSKRIBUS FÜR ANFÄNGER:INNEN



<https://www.oeaw.ac.at/de/acdh/newsevents/event-reihen/acdh-ch-tool-galleries>

The ACDH-CH Tool Galleries, which take place **three times a year**, are workshops dedicated to providing information and **hands-on training in the use of specific digital humanities tools**. Participants are given the opportunity to engage with the tools and to learn from experts with first hand experience in the application of digital methods in the field of humanities research. They are a service **open to everyone** but directed, in particular, at humanities scholars who would like to learn about and engage with computer-based methods in order to implement them in their own research.

Summer Schools

- European Summer University in Digital Humanities „Culture & Technology“
 - <https://esu-ct.conference.ubbcluj.ro/about/>
- Edirom Summer School
 - <https://ess.uni-paderborn.de/>
- Helsinki Digital Humanities Hackathon (DHH)
 - <https://www.helsinki.fi/en/digital-humanities/helsinki-digital-humanities-hackathon-2025-dhh25>
- Digital Humanities @Oxford Summer School (DHOX)
 - <https://digital.humanities.ox.ac.uk/digital-humanities-oxford-summer-school>

→ (application for) financial support possible in most cases

INTERNSHIPS

In February and July the ACDH-CH offers internships for students, academics and young professionals interested in getting to know the institute from the inside.

WHAT WE OFFER

Internships at the ACDH-CH aim to provide a hands-on work experience and an insight into the workings of an academic research institute with a focus on digital humanities scholarship. We are pleased to report that we have had 10 to 40 interns every year since the programme's commencement in summer 2017.

Interns will gain a first-hand impression of the academic world and an understanding of what a career as a DH researcher may entail. In assigning interns to specific research projects or work groups, we will take into account their interests and allow them to practice specific skills, while also giving them the opportunity to expand their academic knowledge and gain new experience.



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ACDH-CH Internships in July 2025

Apply now!

Deadline:
4 May 2025

<https://www.oeaw.ac.at/acdh/education/acdh-ch-internships>



Auf dem Weg nach Hause: Anna, meiner minderjährigen Töchter: „Mama, Bibi hat dich in ihre Snapchatstory aufgenommen – du bist Interview Nummer sieben.“
Ich bin Interview Nummer sieben! In Bibis Windschatten habe auch ich es unter die Top Ten geschafft.

„Natürlich ist
und hörte, w
ße des Wett
abfiel. Er sta
laja, der Tag
klarsachte er fröhlich in die
ging in sein Zimmer.

Staying up to date

Websites

- DHd blog: <https://dhd-blog.org/>
- dha website: <https://digital-humanities.at>
- H-Net (Humanities and Social Sciences Online): <https://networks.h-net.org/> (identify specific networks interesting to you)

Mailing lists

DHd-Mailingliste: <https://lists.uni-hamburg.de/mailman/listinfo/dhd>

The image features a dark background with various abstract elements. On the left, there are two white zigzag lines above a light orange semi-circle. A large, dark rectangle with a thin white border and a light orange drop shadow is centered. The word "Explore!" is written in white inside this rectangle. On the left side of the rectangle, there is a light green circle. At the bottom right, there is a light green semi-circle.

Explore!

Time plan for tool exploration

- (1) Selection of a tool unfamiliar to you (5-10 min.)
- (2) Exploration of tool (30 min.)
- (3) a) Exploration of another new tool (15 min.)
b) Further exploration and discussion together with colleagues interested in the same (group of) tool(s) (15 min.)
- (4) Plenum discussion on experiences, potentials & limitations (15 min.)